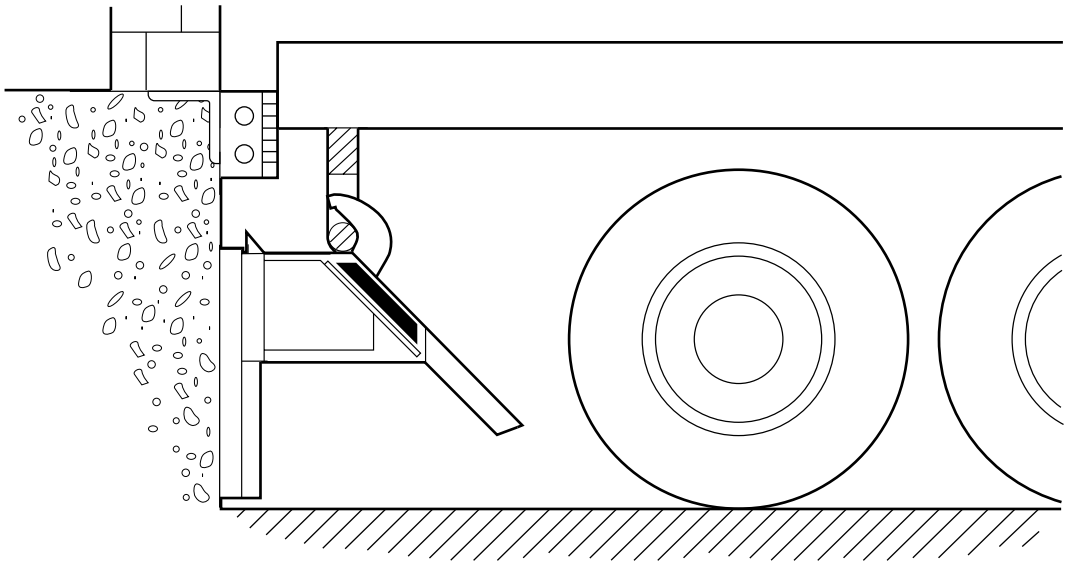
Model LDL-500
DOK-LOK®



RITE HITE®

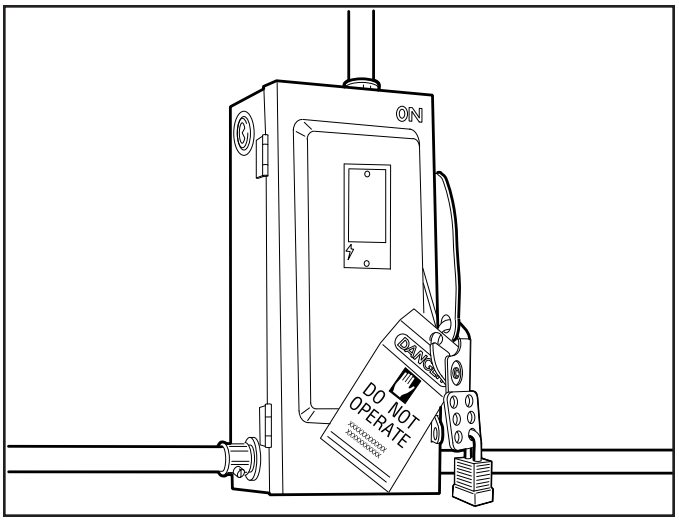
SERIAL NO. DL001000 AND UP
Including 9" Carriage Option

Owner's Manual

Printed in U.S.A.
Copyright 1999

Publication No. 1139-R3
March 1999

SAFETY WARNINGS



⚠ WARNING

When working with electrical or electronic controls, make sure that the power source has been locked out and tagged according to OSHA regulations and approved local electrical codes.

LOCKOUT/TAGOUT PROCEDURES

The Occupational Safety and Health Administration (OSHA) requires, in addition to posting safety warnings and barricading the work area (including, but not limited to, trucking office and loading docks), that the power supply has been locked in the OFF position or disconnected. It is mandatory that an approved lockout device is utilized. An example of a lockout device is illustrated. The proper lockout procedure requires that the person responsible for the repairs is the only person who has the ability to remove the lockout device.

In addition to the lockout device, it is also a requirement to tag the power control in a manner that will clearly note that repairs are under way and state who is responsible for the lockout condition. Tagout devices have to be constructed and printed so that exposure to weather conditions, or wet and damp locations, will not cause the tag to deteriorate or become unreadable.

RITE-HITE® does not recommend any particular lockout device, but recommends the utilization of an OSHA approved device (refer to OSHA regulation 1910.147). RITE-HITE® also recommends the review and implementation of an entire safety program for the Control of Hazardous Energy (Lockout/Tagout). These regulations are available through OSHA publication 3120.

⚠ DANGER

This is the highest level statement. Failure to follow the listed instructions will most likely result in severe injury or death.

⚠ CAUTION

The statements used with this level of warning deal with a safe operating procedure. If the procedure is ignored, the possibility of personal injury may exist.

⚠ WARNING

This is a statement of serious hazard. Failure to follow the listed instructions could place the individual at risk of serious injury or death.

IMPORTANT

IMPORTANT is used to draw attention to a procedure that needs to be followed to prevent machine damage.

TABLE OF CONTENT

INTRODUCTION i
DEFINITION AND FUNCTION 1
INSTALLATION/APPLICATION INFORMATION 5
WIRING DIAGRAM 12
OPERATING PROCEDURE 13
SUGGESTED DOK-LOK® MAINTENANCE 17
TROUBLESHOOTING 22
REPLACEMENT PARTS 26

INTRODUCTION

Read and understand this manual before attempting to install or operate any DOK-LOK®. For the best results, have this product serviced by your authorized RITE-HITE® representative.

The LDL-500 DOK-LOK® by RITE-HITE® is a new concept in dock safety equipment. It is intended to provide a safer work place for workers in shipping and receiving dock areas. The LDL-500 DOK-LOK® is an electro-mechanical restraint device that, when properly installed and operated, retains a secure connection between the truck and dock. An electric motor rotates a heavy-duty hook upward, locking the truck's Rear Impact Guard (R.I.G.) to the dock face. Signal lights, warning horn and signs provide instructions to the truck driver and DOK-LOK® operator that a safe condition exists. The DOK-LOK® is operated from an inside control panel by pressing a control box button.

NOTICE TO USER

Your local RITE-HITE® representative provides the Planned Maintenance Program (P.M.P.) which can be fitted to your specific operation. Call your local representative or RITE-HITE® at 414-355-2600.

If any procedures for the installation, operation, or maintenance of the LDL-500 DOK-LOK® have been left out of this manual or are not complete, contact RITE-HITE®.

The RITE-HITE® products in this manual are covered by one or more of the following U.S. patents: 4,264,259; 4,373,847; 4,379,354; 4,443,150; 4,472,099; 4,560,315 (RE: 32,968); 4,605,353 (RE: 33,154); 4,634,334; 4,692,755; 4,843,373; 5,212,846; 5,297,921; 5,375,965; 5,453,735; 5,531,557; 5,553,987; 5,582,498; 5,664,930; 5,702,223; 5,762,459; and pending U.S. and foreign patent applications. RITE-HITE®, GUARDRITE®, SAFE-T-LIP®, HYDRACHEK®, WHEEL-LOK™, DOK-LOK®, DUAL-DOK®, and SAFE-T-GATE™ are trademarks of RITE-HITE®.

DEFINITION AND FUNCTION

The Model LDL-500 DOK-LOK® is an electro-mechanical, self-aligning restraint device used to secure trucks and semi-trailers (with an intact Rear Impact Guard [R.I.G.] Assembly) to the face of a loading dock. This is achieved by hooking the R.I.G. Assembly with an electrically powered steel hook. This prevents forward movement of the truck/trailer that may create an unsafe void between the face of the dock and the rear end of the truck/trailer as a forklift travels from the loading dock onto the trailer; or to create an obstruction noticeable to the truck driver, should the driver accidentally try to pull the truck/trailer away while it is being serviced.

The proper or improper activation of the hook is monitored by:

VISUAL CONTROL – one set of flashing green or red lights located at the inside of the building for the forklift operator, and one set located outside of the building for the truck driver.

AUDIO CONTROL – a horn will sound at the inside of the building, warning the forklift operator if there is no R.I.G. present, or if the engagement is improper. In this case, the trailer must be secured by other means (wheel chocks, etc.) prior to servicing trailer.

Prerequisite for proper hook engagement is that the trailer is parked firmly against a 4" (trade standard) thick dock bumper. The activation/deactivation is solely controlled from the inside of the building by momentarily depressing either the LOCK (raise) button or the UNLOCK (lower) button.

NOTE

There is no need to hold in either button to maintain its function. Once a button is depressed, the hook will travel automatically for a preset time interval.

The normal mode of the hook is in the lower stored position, showing a flashing red light (trailer not secured) at the inside of the building and a flashing green light (trailer free to move to or away from the loading dock) at the outside of the building.

In addition to the lights, there are three instruction signs. The smaller sign is mounted over the DOK-LOK® control box. The sign reads: “CAUTION: ENTER ON GREEN LIGHT ONLY!” There are two large outside signs, one in straight print and one in mirror-image print. These signs are mounted near the outside lights, the signs read: “CAUTION: BACK IN OR PULL OUT ON GREEN LIGHT ONLY!”.

Once the trailer is parked, the dock attendant will depress the LOCK button. This will raise the hook to engage the R.I.G. As soon as the R.I.G. is properly locked, there will be simultaneous light change — the inside will change from red to green flashing (ready for service) and the outside will change from green to red flashing (don’t pull out). After the service is completed, the dock attendant will have to depress the UNLOCK button which then will return the hook to its lower stored position.

A proper hook engagement is achieved when the hook raises unobstructed to fully trap the horizontal cross-member of the R.I.G. Assembly. An improper hook engagement is if the horizontal crossmember of the R.I.G. is missing or it is bent or located so far toward the rear axle of the trailer that it will prevent the free passage of the hook. In either case, the lights will stay in a non-serviceable mode and a horn will sound. At this point, the trailer must be secured by other means (example: wheel chocks) in order to become serviceable.

NOTE

On occasion the truck may move forward several inches, trapping the R.I.G. This is still a safe “lock” or operating position, but will require the driver to back the trailer up slightly to enable the hook to release.

FEATURES

See Figure 1 for locations of these features:

R.I.G.

R.I.G. is an acronym used for a “Rear Impact Guard” that is Federally mandated to be located on the rear of over the road trailers to prevent accidental underride by automobiles.

ROLLER TRACK ASSEMBLY

This assembly is mounted to the loading dock wall to guide the CARRIAGE ASSEMBLY in a vertical plane and provide the top mount for the CARRIAGE RETURN SPRINGS. Incorporated features in this assembly are SPRING GUARDS, SPRING COVER STOPS and a CARRIAGE ASSEMBLY TOP STOP. It also transmits the creep force or pull out force exerted on the HOOK from a trailer to the loading dock wall.

CARRIAGE ROLLER ASSEMBLY

This roller is comprised of an outer housing with a needle bearing that is pre-lubricated to allow easy movement of the CARRIAGE ASSEMBLY and retain the CARRIAGE ASSEMBLY within the ROLLER TRACK ASSEMBLY.

CARRIAGE RETURN SPRINGS

These springs bias the CARRIAGE ASSEMBLY to its stored full up position within the ROLLER TRACK ASSEMBLY or to maintain contact with the R.I.G. (Rear Impact Guard) of a trailer.

LOWER SPRING BAR

This bar provides the bottom mount for the CARRIAGE RETURN SPRINGS and the CARRIAGE ASSEMBLY TOP STOP.

CARRIAGE ASSEMBLY

The CARRIAGE ASSEMBLY, with its triangular housing and sloped front, is constructed of abrasion resistant materials designed to be contacted by the R.I.G. of a trailer. This allows the CARRIAGE ASSEMBLY to be self-positioning to restrain the trailer once the HOOK has been activated by the dock attendant.

CARRIAGE ASSEMBLY SLOPE EXTENSION

This allows trailers with low ground clearance R.I.G.s to be serviced by the DOK-LOK® and is constructed of high grade steel tubing to resist abrasion.

HOOK

The HOOK is housed within the CARRIAGE ASSEMBLY and provides an obstruction in front of the trailer R.I.G. to prevent it from rolling/creeping away from the dock. The HOOK is constructed from a high grade steel to provide a long service life. The HOOK is put into position by the MOTOR ASSEMBLY.

MOTOR ASSEMBLY

This provides the means of moving the HOOK to its active position.

SPRING COVERS AND INTEGRAL LIP GUIDE

There are two independent SPRING COVERS, the right hand SPRING COVER having an INTEGRAL LIP GUIDE which is used to guide the lip past the ROLLER TRACK ASSEMBLY in a below dock end load condition. These SPRING COVERS protect the springs and keep debris out of the ROLLER TRACK ASSEMBLY when the LDL-500 is in its stored full up position.

LIMIT SWITCHES, CONTROL HARNESS, CONTROL BOX, OUTSIDE LIGHT BOX, AND SIGNAGE

The combination of these components is used to interface the LDL-500 vehicle restraint to the dock attendant, providing a means of control to operate the LDL-500 visual and audio communications, as well as providing a visual communication to the trailer driver.

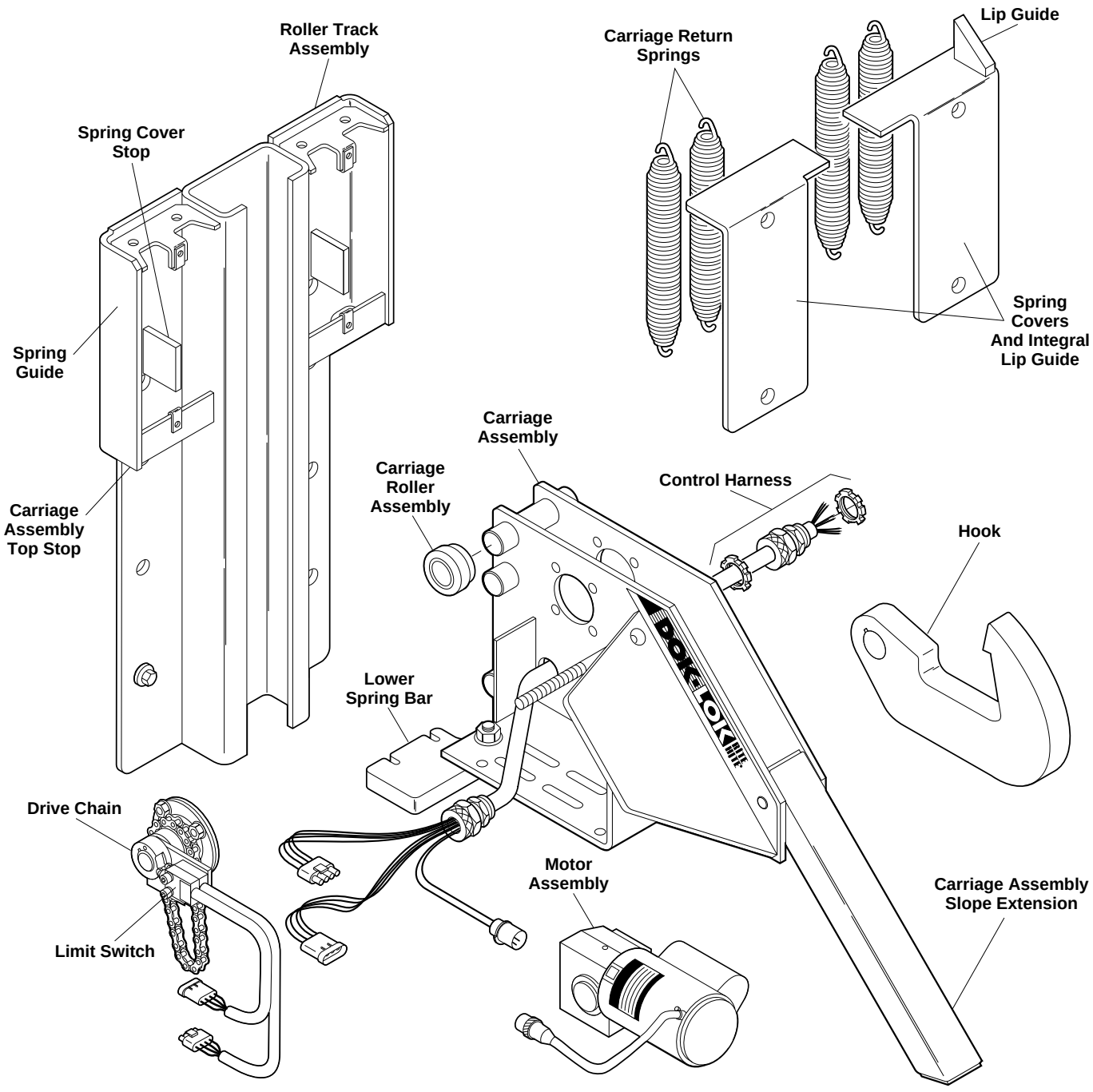
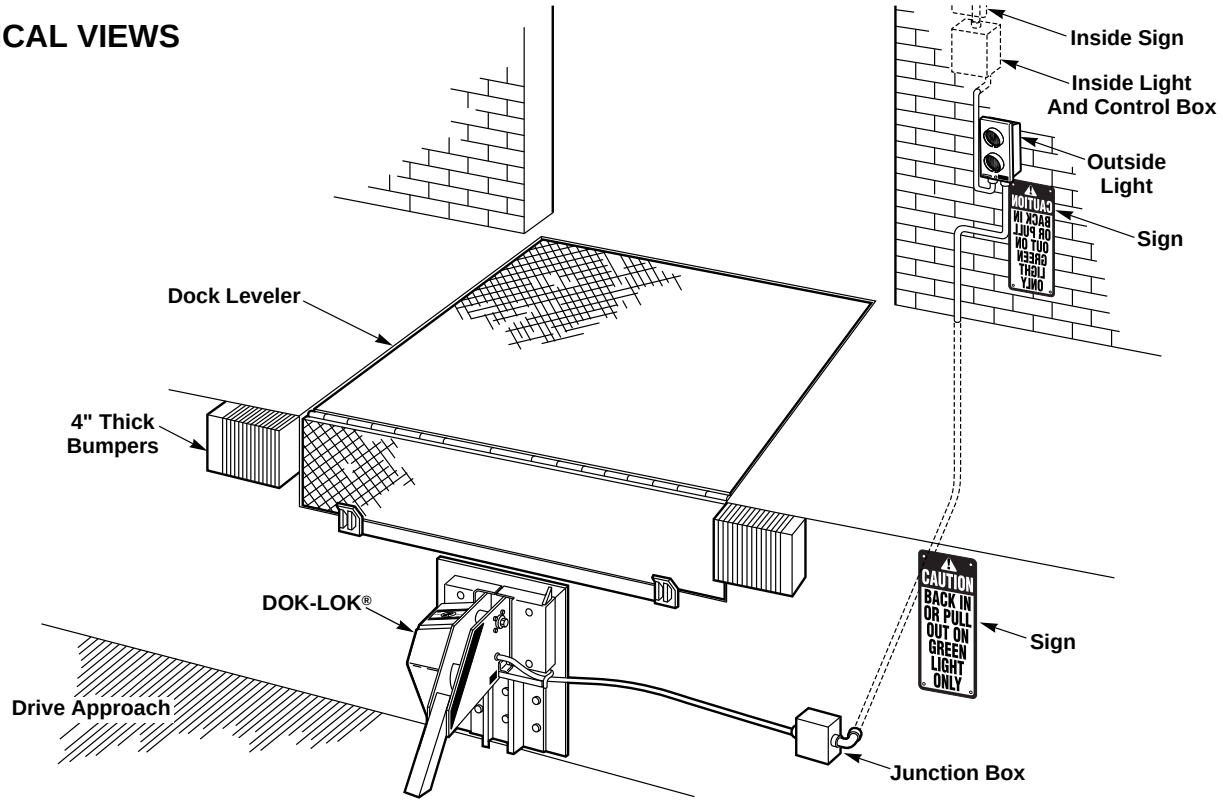
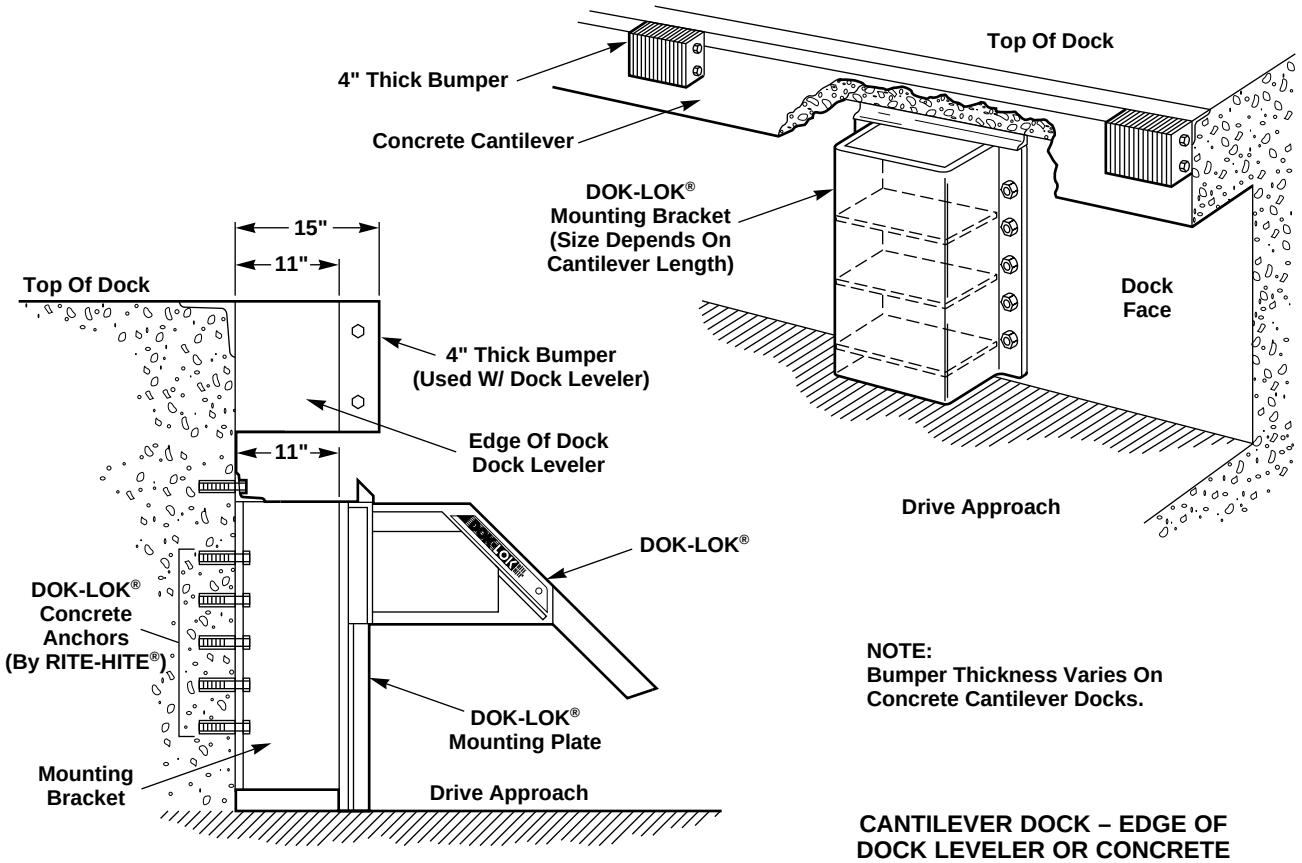


FIGURE 1 — LDL-500 DOK-LOK® FEATURES

TYPICAL VIEWS



LEVELER IN A RECESSED PIT



NOTE:
Bumper Thickness Varies On Concrete Cantilever Docks.

CANTILEVER DOCK – EDGE OF DOCK LEVELER OR CONCRETE

INSTALLATION/APPLICATION INFORMATION

The performance of any machine can be no better than the installation and application. The DOK-LOK® requires only ordinary application and installation skills using good workmanship and quality materials.

You may install a DOK-LOK® on docks with RITE-HITE® or other levelers already installed in recessed pits or mounted on the top edge of dock. You may install a DOK-LOK® on docks without any sort of leveler.

IMPORTANT

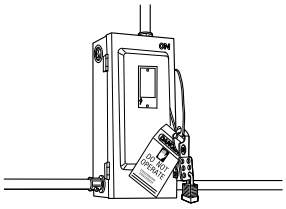
Whether there is a leveler or not, a 4 inch thick bumper is required. DO NOT install a DOK-LOK® on docks without 4 inch thick bumpers. For thicker bumpers consult factory.

Follow the 7 step Installation Procedure. These are:

1. Inspect DOK-LOK® materials
2. Install roller track plate
3. Install DOK-LOK® to roller track plate
4. Install electrical components
5. Install signs
6. Lubricate
7. Test Operation

DOK-LOK® POWER SOURCE

⚠ WARNING



LOCKOUT/TAGOUT PROCEDURES

When working with electrical or electronic controls make sure that the power source has been locked out and tagged according to OSHA regulation 1910.147

On occasion, the incoming electrical power for the LDL-500 DOK-LOK® is taken from a convenient nearby electrical appliance, i.e., overhead door opener. Generally, there is no problem using this sort of power source; you must, however, verify that the amperage is in accordance with local and federal codes.

The DOK-LOK® vehicle restraint requires 120V., single phase power 60 Hz. power and 10 amps of current, which is from a 1/4 HP motor, to operate properly.

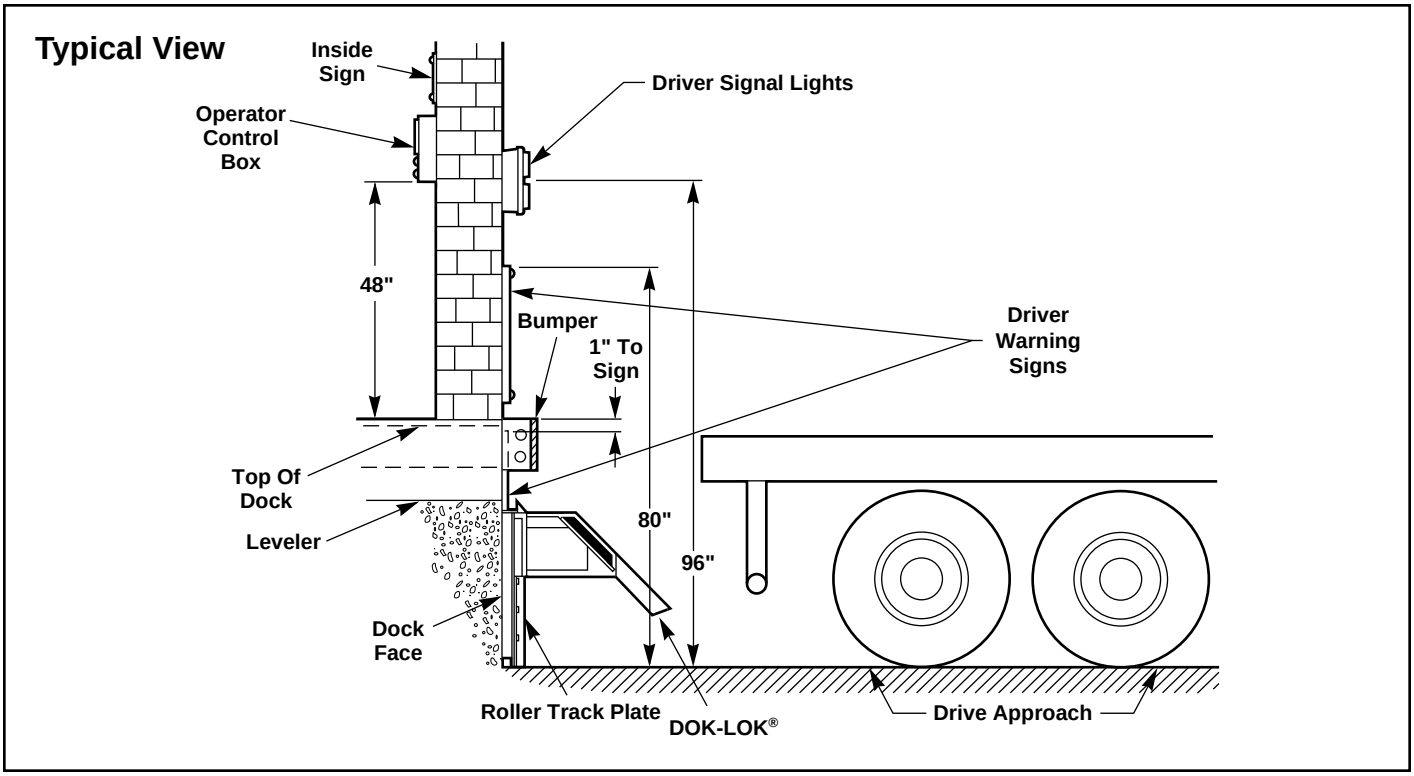
Two (2) DOK-LOK® vehicle restraints can be hooked into one (1) 20 amp branch circuit breaker per the 1996 National Electrical code Paragraph #430-53.

When in doubt, call your RITE-HITE® representative or RITE-HITE® at 414-355-2600.

All components must be connected to a SAFETY EARTH GROUND where indicated. The SAFETY EARTH GROUND must conform to the 1996 National Electric Code Section 250-81 section (a) or section (c) for a grounding electrode system.

DOK-LOK® EMBEDDED MOUNTING PLATE

On new or remodeled pit construction, using the combined front pit curb angle and DOK-LOK® embedded mounting plate makes good sense. The 23-1/2" x 20" plate is welded to the front curb angle. Both have concrete anchors. Contractors find it an easy method of forming the front dock face and installers prefer to weld on the DOK-LOK® roller track plate as opposed to drilling and setting concrete anchors.



INSPECT DOK-LOK® MATERIALS

Open and inspect all materials. Immediately report any damage or material shortage. Study component assemblies and determine their proper locations.

INSTALL ROLLER TRACK PLATE

The LDL-500 DOK-LOK® is shipped assembled. The track roller plate must be disassembled from the DOK-LOK® to begin installation. To disassemble,

1. Remove spring cover and springs. See Figure 2.
2. Remove motor cover.
3. Remove nuts from spring bar.
4. Roll DOK-LOK® out of roller track plate. Refer to Figure 3.

IMPORTANT

Be careful to keep rollers on the axles.

Install roller track plate onto dock face at specified location by welding to an embedded steel plate or by using the 15 concrete anchors provided in conjunction with welding to pit steel and a leveler frame. Refer to Figures 4 and 5.

NOTE

Trim roller track plate, as necessary, up to a maximum of 4" from the bottom. Refer to RITE-HITE® Application Guidelines.

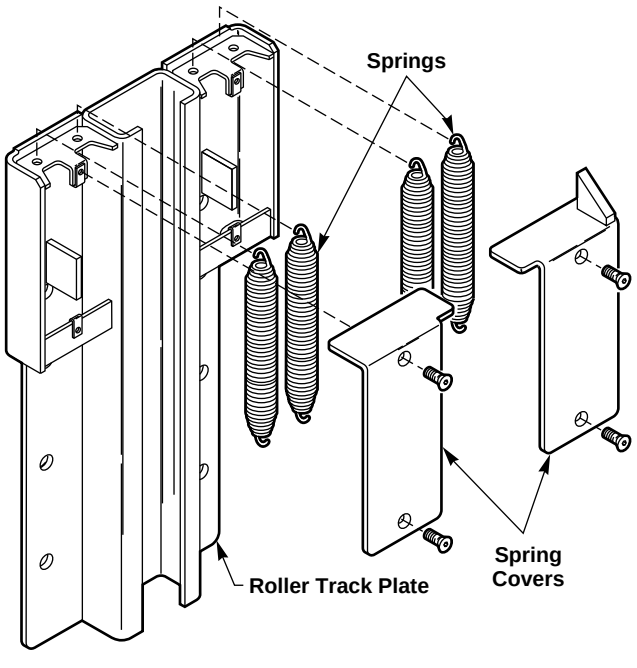


FIGURE 2 — ROLLER TRACK ASSEMBLY

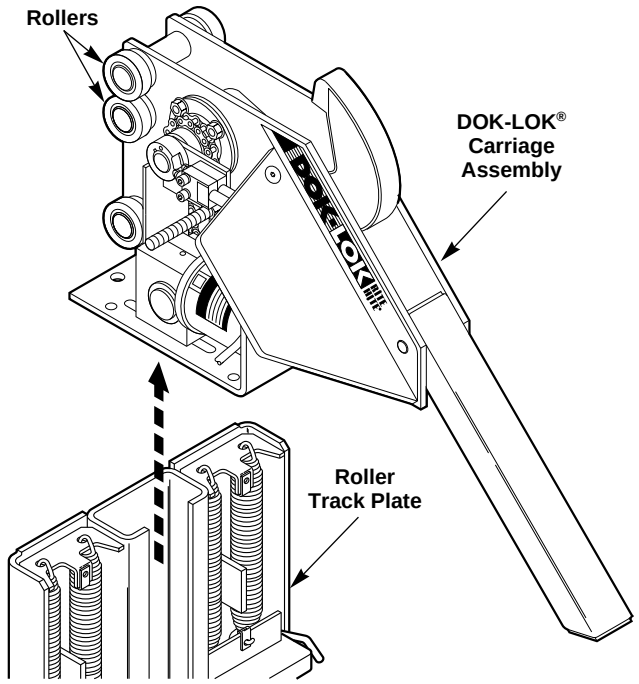


FIGURE 3 — REMOVING CARRIAGE FROM TRACK

Welding Installation Instruction

IMPORTANT

DOK-LOK® Welding

Never weld on the LDL DOK-LOK® after motor is wired into the control box and the power to the box is on. Electrical current from the welder can "feedback" through the circuit and damage the motor and other components.

If the installation being worked on is a "reinstallation" the following electrical connections must be disconnected prior to welding. Make sure to Lock Out and Tag Out the power at the fused disconnect, then, remove motor and limit switch connections from the control harness located inside the motor cover and disconnect the ground located in the outside junction box. Once all welding procedures have been completed, reconnect all wires.

IMPORTANT

A plumb roller track plate is required. If the dock face is not plumb/perpendicular, contact your RITE-HITE® representative or RITE-HITE® at 414-355-2600.

The carriage roller track cannot be bent or deformed. Straighten or replace as required.

NOTE

Some mechanical dock levelers have an adjusting nut access hole in the leveler front subframe. If the LDL-500 DOK-LOK® roller track interferes with the access hole, the track plate must be cut to allow access. Refer to Figure 6.

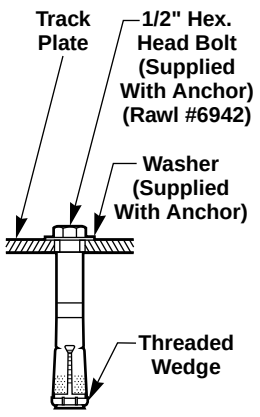
Some levelers may be slightly recessed within pit. A shim must be inserted between the roller track plate and the leveler front subframe and then must be welded in place. Refer to Figure 7 for adding a necessary "lip guide."

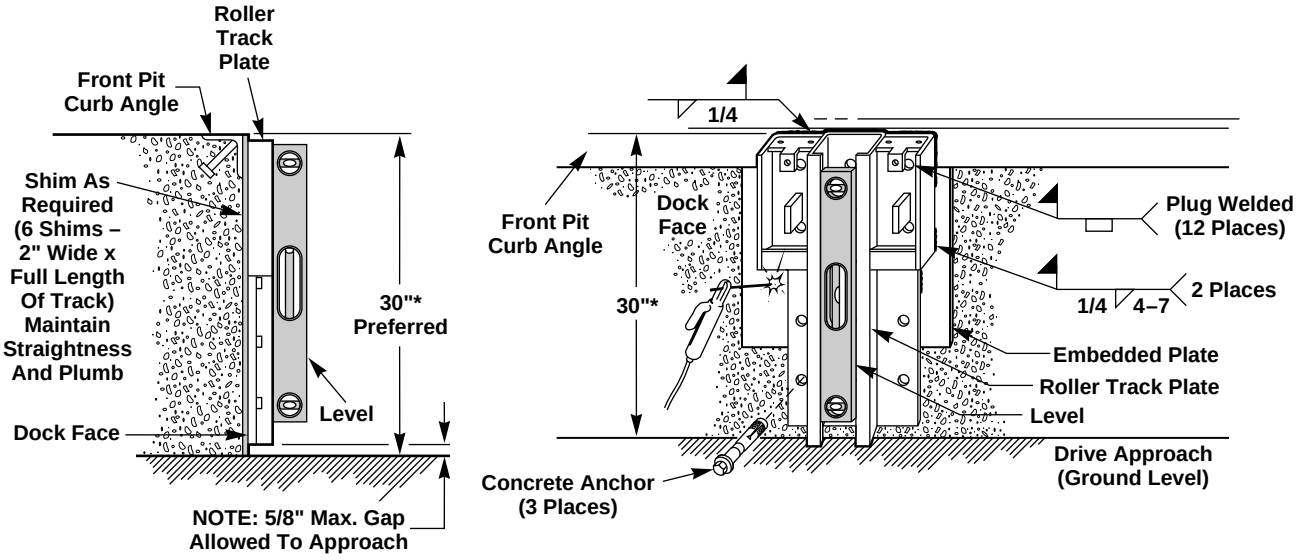
Anchor Installation Instructions

IMPORTANT

Fifteen (15) concrete anchors are provided with each DOK-LOK®. You must install an anchor in each roller track plate hole except those plug-welded to embedded steel.

1. Put roller track plate in place.
2. Drill hole of 5/8" diameter and minimum of 4-5/8" deep. Clean out hole. Remove inspection tag and do not expand before installation. Insert anchor and drive flush with roller track plate, making certain that the threaded wedge is put in first. Do not disassemble anchor prior to installation. Install all anchors and torque to 100 ft-lbs.





NOTES: Shims Must Be The Full Length Of The Roller Track Mounting Plate. Minimum Electrode Must Be 1/8" 7018 Or Better.

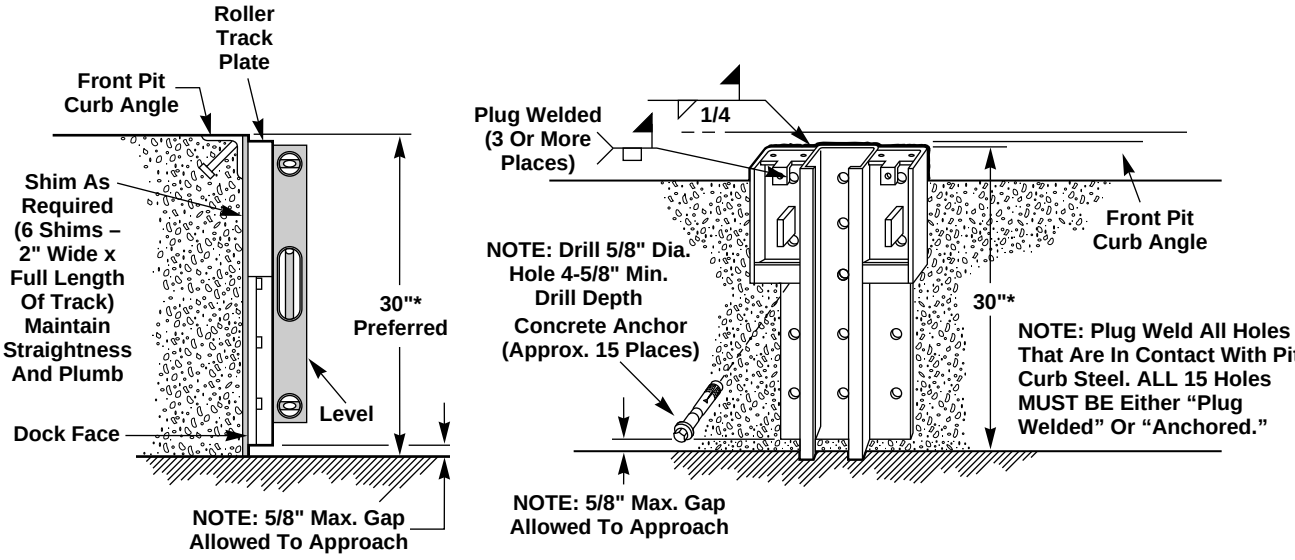
Never Install DOK-LOK® Directly Onto Concrete Block Or Brick Dock Face.

When Welding DOK-LOK® Disconnect Power And Ground Leads To DOK-LOK® And Leveler.

IMPORTANT: DO NOT Bend Or Deform Roller Track. Be Sure To Mount Track Roller Plate Vertically Plumb On Dock Face.

*** Due To Actual Dock Conditions, Total Mounting Height May Be Different. Refer To RITE-HITE® Application Guidelines For Actual Mounting Height.**

FIGURE 4 — WELDING INSTALLATION METHOD



NOTES: Shims Must Be The Full Length Of The Roller Track Mounting Plate. Minimum Electrode Must Be 1/8" 7018 Or Better.

Never Install DOK-LOK® Directly Onto Concrete Block Or Brick Dock Face.

When Welding DOK-LOK® Disconnect Power And Ground Leads To DOK-LOK® And Leveler.

IMPORTANT: DO NOT Bend Or Deform Roller Track. Be Sure To Mount Track Roller Plate Vertically Plumb On Dock Face.

*** Due To Actual Dock Conditions, Total Mounting Height May Be Different. Refer To RITE-HITE® Application Guidelines For Actual Mounting Height.**

FIGURE 5 — ANCHOR INSTALLATION METHOD

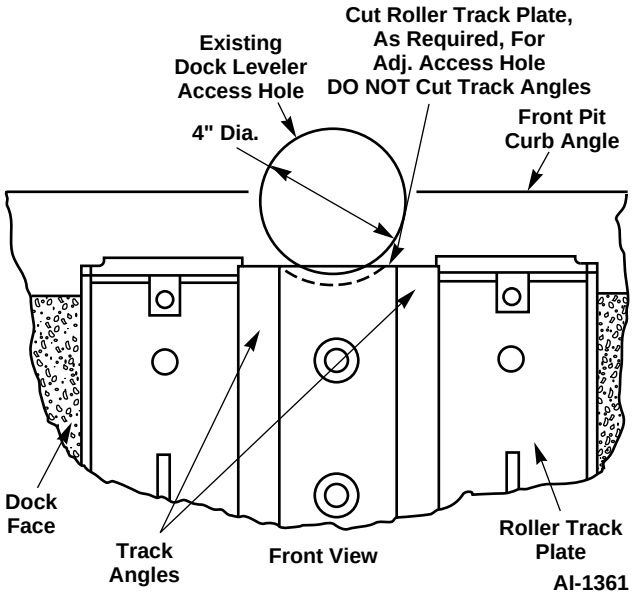


FIGURE 6 — ROLLER TRACK — FRONT VIEW

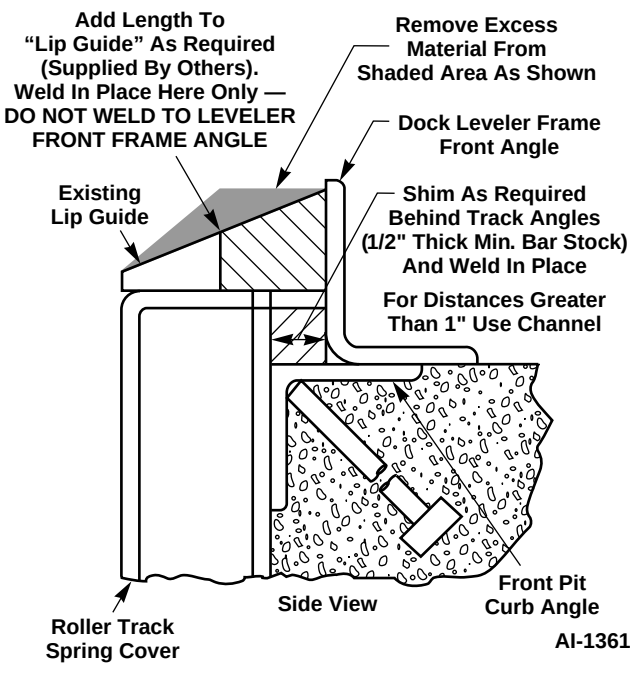


FIGURE 7 — ROLLER TRACK — SIDE VIEW

INSTALL DOK-LOK® TO ROLLER TRACK PLATE

Attach the 4 springs to the lower spring bar. Pull upward and slide over top spring mount on the track roller plate. (These parts were removed in Step 2.) Then slide the carriage assembly into the roller track. Position and bolt the lower spring bar to the bottom of the DOK-LOK® carriage. Install the spring covers and motor cover. Refer to Figure 8.

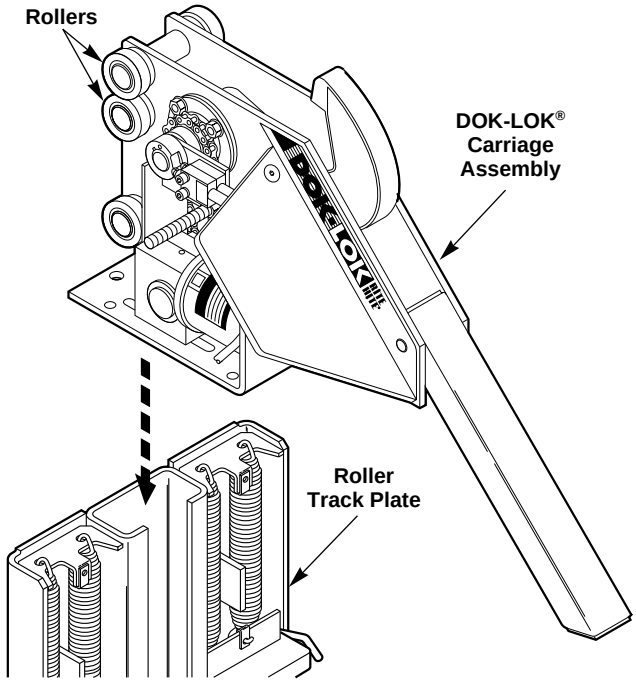


FIGURE 8 — INSTALLING CARTRIDGE TO ROLLER TRACK

INSTALL ELECTRICAL COMPONENTS

The DOK-LOK® Assembly includes a 63" long flexible wiring harness. Also provided are the control box with lights, and the outside signal light box. The outside junction box, conduit fittings and wire are provided by others. Quality material and workmanship should be used. For installation refer to Figures 9, 10 and the Wiring Diagram on page 12.

INSTALLATION OF SIGNS

There are two Driver Warning Signs and one inside operator sign supplied with each DOK-LOK®. Because of the many ways signs can be mounted, hardware is not supplied. Mount both outside signs on the right hand side of the DOK-LOK®. The “mirror-image” sign is usually mounted above the lower sign and near the outside signal lights. (“Mirror-image” permits the truck driver to read sign from truck mirror.) Mount the inside sign next to the control box. All signs must be installed.

Figure 10 provides a suggested location for sign and outside light mounting. Good workmanship and quality material should be used.

NOTE

Some docks may have dock/truck seals (or shelters) installed or planned which are larger than standard. Outside light, sign location, and mounting should be studied before proceeding to avoid interference.

If required for “special reasons,” signs may be trimmed for fit. NEVER cut or eliminate sign letters/words.

NEVER put conduit in front of signs. A clear view of both exterior signs and the single interior sign must be maintained at all times.

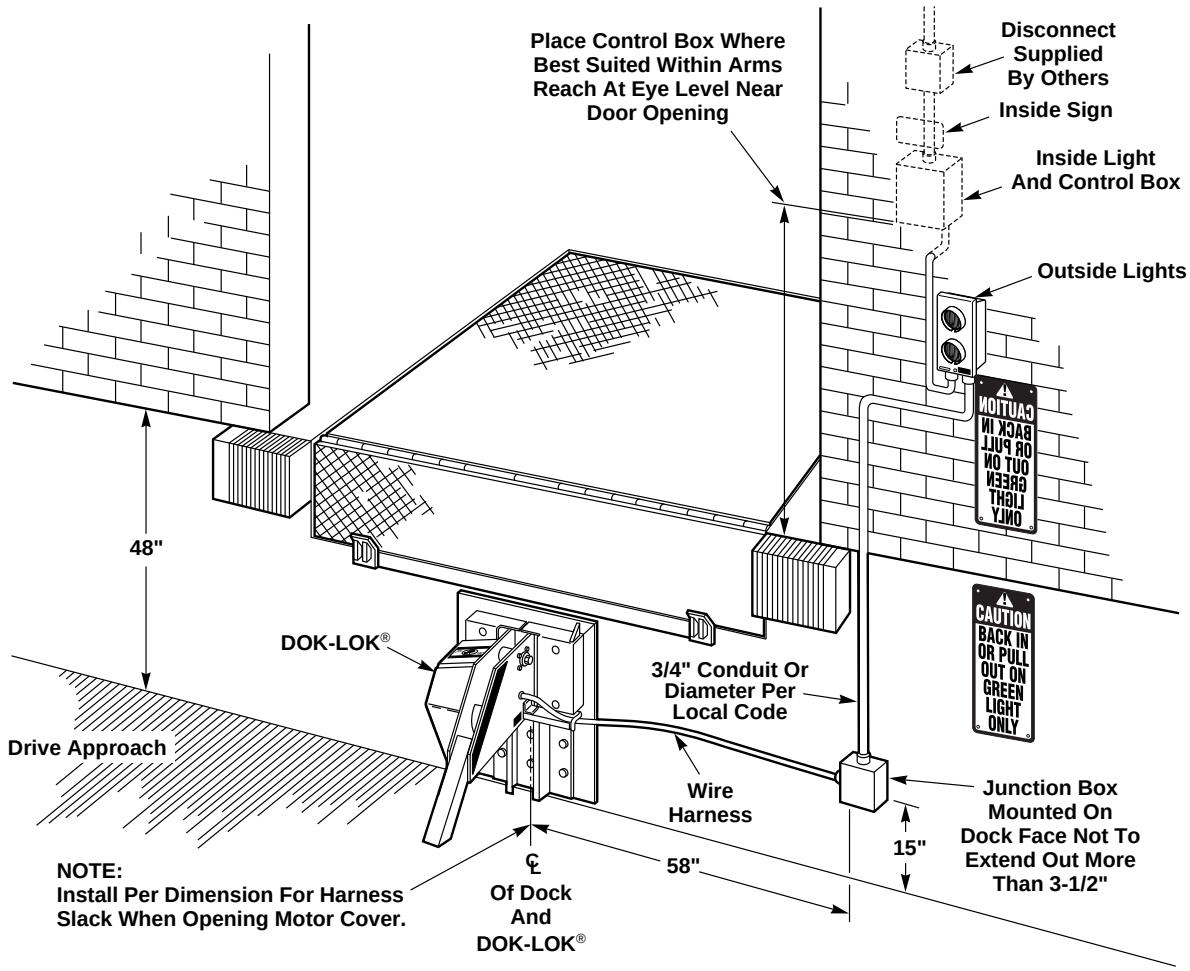


FIGURE 9 — ELECTRICAL INSTALLATION LAYOUT

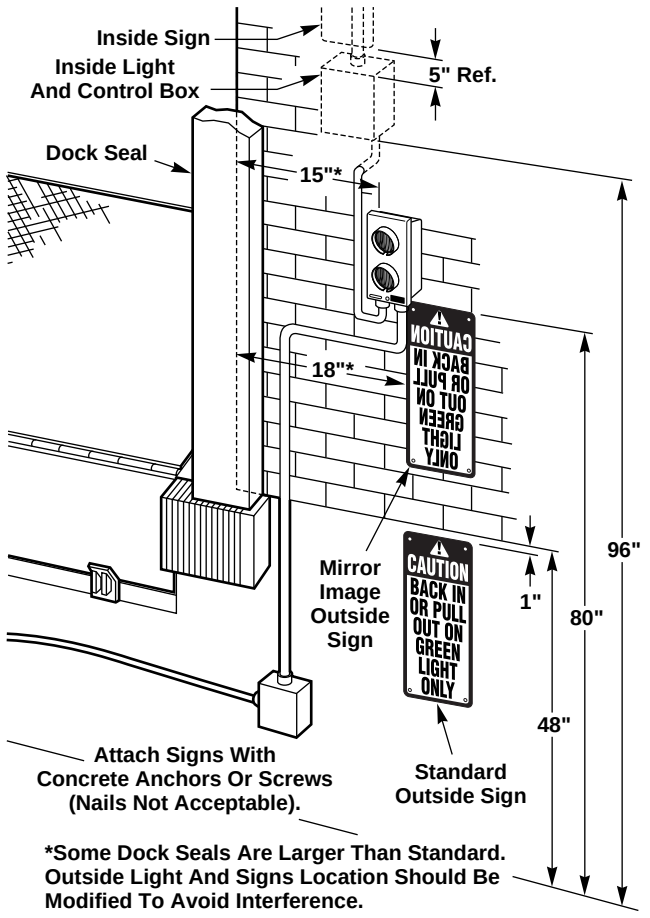


FIGURE 10 — SUGGESTED SIGN LAYOUT

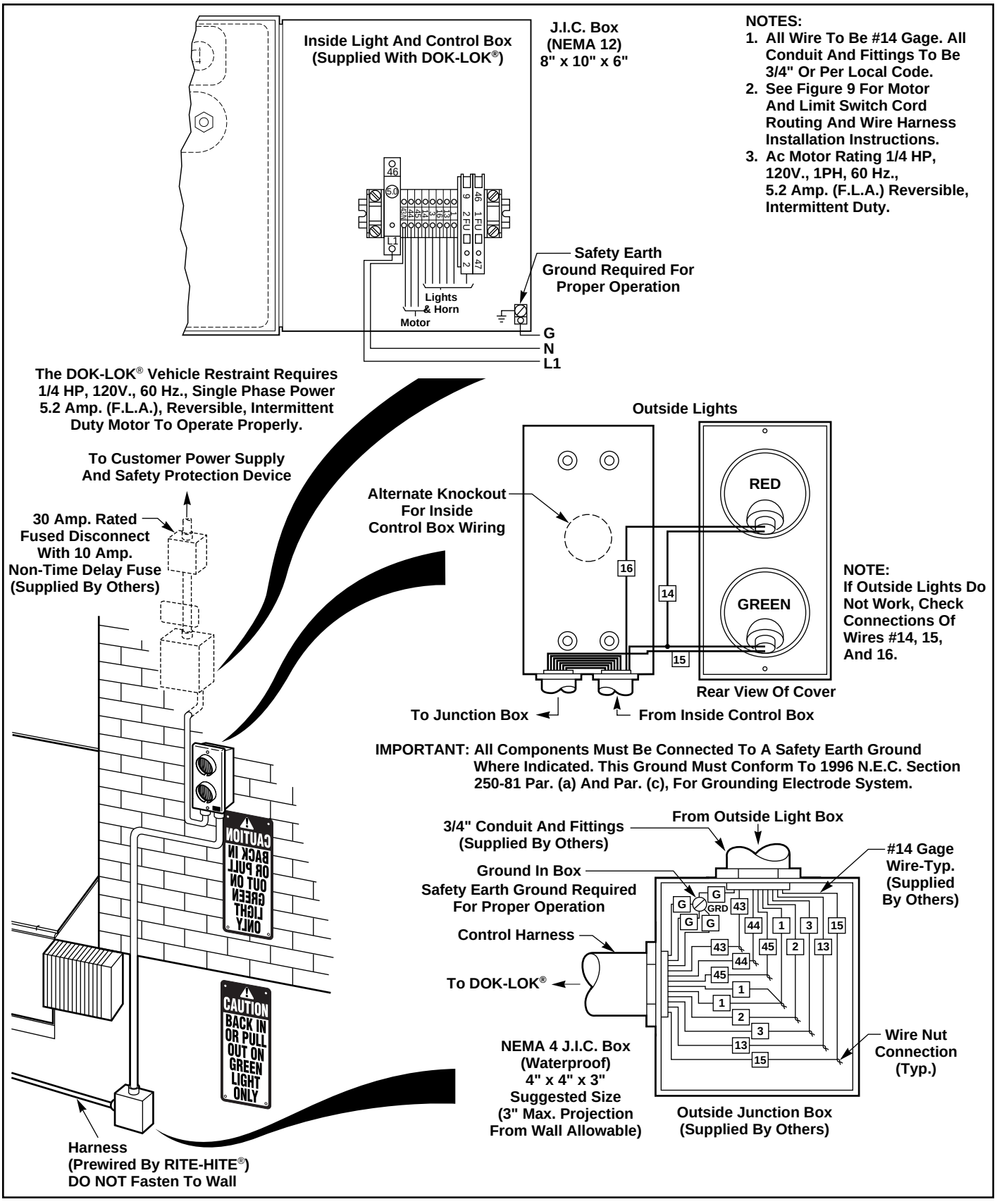
LUBRICATE

If a unit is being relocated, follow lubrication instructions in the Maintenance Section, page 17. If a new installation, there is no need, since it is fully lubricated during assembly at factory.

OPERATING TEST

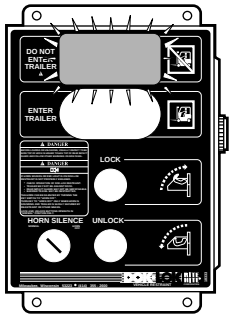
Follow the Operating Procedure beginning on page 13.

WIRING DIAGRAM



OPERATING PROCEDURE

INSIDE AND OUTSIDE “GREEN-RED” LIGHT SEQUENCE

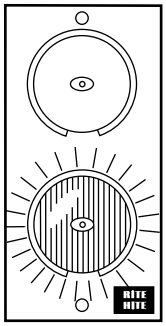


Inside Lights

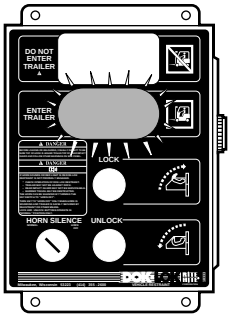
1. **Without Trailer At Dock**

   Inside Red Light Flashing (Warning to forklift operator)

   Outside Green Light Flashing (Telling truck driver to back into dock or pull out)



Outside Lights

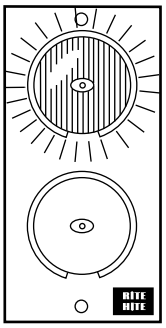


Inside Lights

2. **With Trailer “Locked” At Dock**

   Inside Green Light Flashing (Telling forklift operator truck is locked at dock and safe to service)

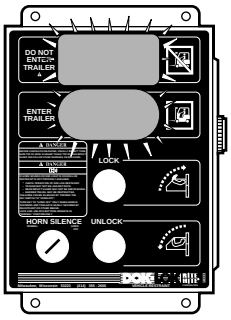
   Outside Red Light Flashing (Warning truck driver his truck is locked at dock and should not pull away)



Outside Lights

WARNING

If the inside red light appears while servicing a truck, press the lock button to reactivate (lift) the hook. If the light does not change to green, the truck should be safely secured via other means. Investigate the problem and make any needed repairs.

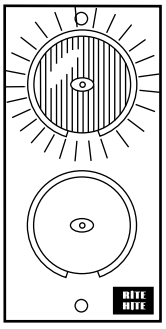


Inside Lights

3. **Horn Silence Mode**

   Inside Alternating Red and Green Flashing (Telling Dock Attendant that truck has been secured by other means)

   Outside Red Light Flashing



Outside Lights

BEFORE ACTIVATING “HORN SILENCE” SECURE TRAILER BY OTHER MEANS.

LOCK AND UNLOCK FUNCTIONS OPERATE WHEN KEY SWITCH IS IN NORMAL POSITION ONLY

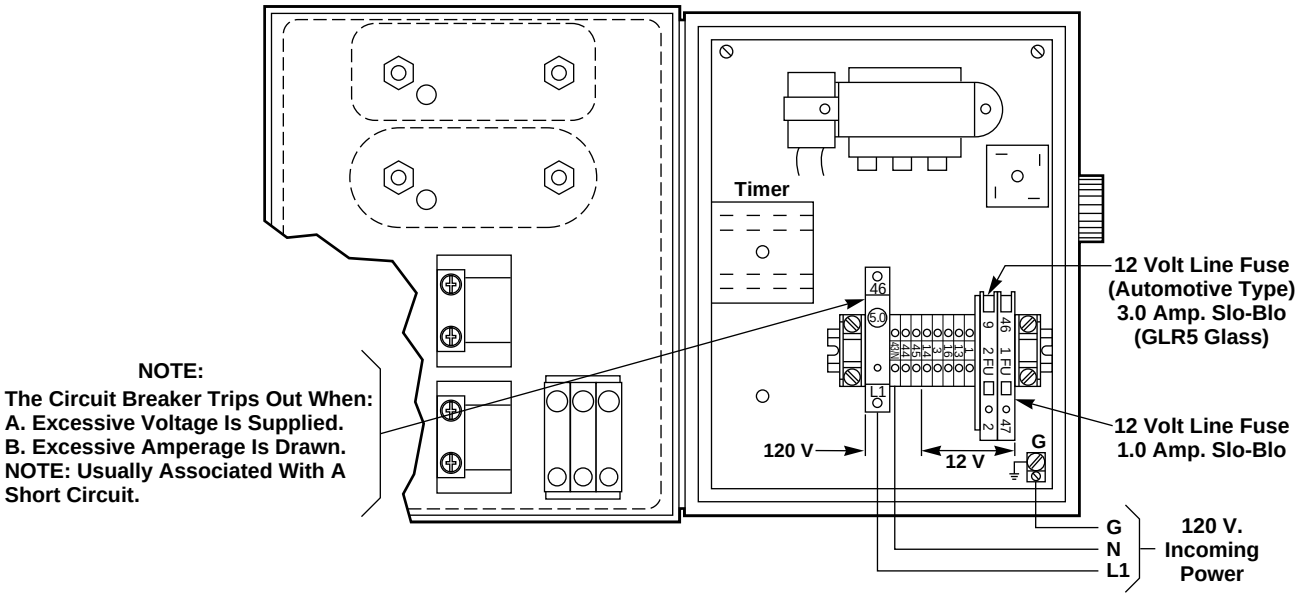


FIGURE 11 — HORN SILENCE/CONTROL BOX

⚠ WARNING

1. **Before loading or unloading a vehicle at your loading dock while using a DOK-LOK® Model LDL-500 vehicle restraint, always visually inspect to be sure that the DOK-LOK® hook entraps the R.I.G. assembly as shown in View A. If a condition occurs like that shown in View C (and it cannot be remedied by backing the trailer firmly against the dock bumpers), secure the trailer by other means.**
2. **Be sure that the area around the R.I.G. assembly is free of plates or other obstructions.**
3. **Always operate the DOK-LOK® vehicle restraint from the top of the dock.**
4. **Inspect all restraint lights and horns daily to make certain they work properly.**
5. **Perform maintenance on restraints in accordance with the Owner's Manual.**
6. **DOK-LOK® vehicle restraints should be operated only by authorized personnel who have read and understand the Owner's Manual.**

Call RITE-HITE® at 414-355-2600 with any questions.

FAILURE TO FOLLOW THESE PROCEDURES COULD ALLOW UNEXPECTED TRAILER/LOADING DOCK SEPARATION RESULTING IN DEATH OR SERIOUS INJURY!

OPERATION

1. To hook trailer press LOCK button. Inside light will turn green. View “A.” If horn sounds, see Step 4.

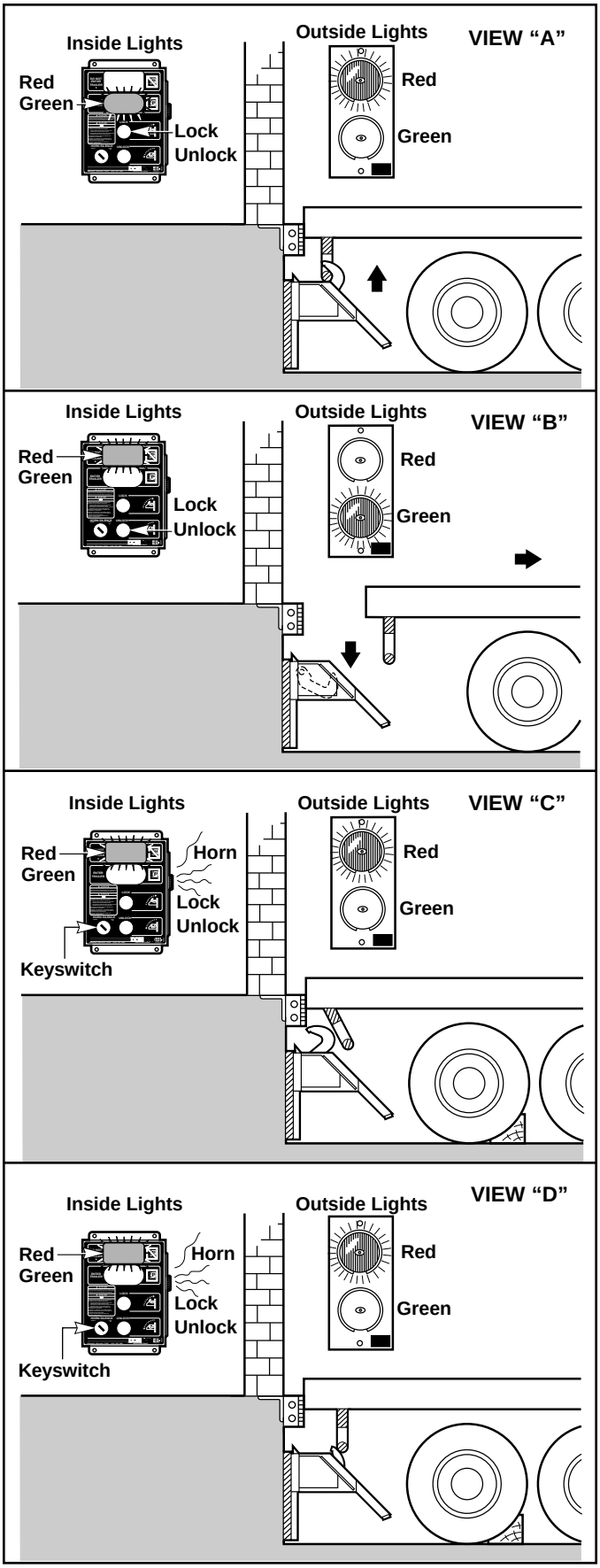
   NOTE

   The truck/trailer must be parked tight against the face of both dock bumpers. Daily inspect and test outside lights, inside lights, and horn.

⚠ WARNING

Always operate the DOK-LOK® from top of dock to prevent personal injury.

2. Inspect to ensure that the DOK-LOK® hook securely entraps the R.I.G. of the vehicle being serviced before operating the dock leveler.
3. To unhook trailer press UNLOCK button. Inside light will turn red. View “B.”
4. If horn sounds during Step 1:
   a. Check if the trailer is parked firmly against the dock bumpers. If not, have trailer backed up and repeat Step 1.
   b. If trailer is parked tight against both dock bumpers and there is no hookup possible, probable cause: R.I.G. cross brace missing, or R.I.G. is bent and/or located too far towards the rear trailer axle. View “C” and “D.”
   c. Secure trailer by other safety means prior to attempting service. The DOK-LOK® inside light will stay flashing RED to remind the forklift operator that there is a potential danger of the trailer being pulled out during the service operation. The outside light will remain RED warning the truck driver not to pull away. The horn will continuously sound until silenced.
   d. To silence horn, turn horn silence keyswitch to HORN OFF position. Once silenced, the LOCK and UNLOCK buttons are disabled, and both the inside red and green lights flash. To unlock the trailer, turn keyswitch to NORMAL position and press the UNLOCK button.
5. Operate dock leveler only when the truck is properly and safely restrained by DOK-LOK® or alternate means.



CIRCUIT BREAKER

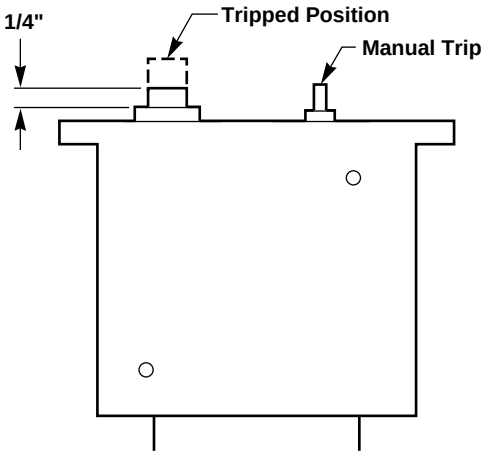
5.0 Amp.
CONTROLS INCOMING
120V. 1PH. POWER
SHORT CIRCUITS AND
MOTOR OVERLOADS

NOTE

The circuit breaker trips out when:

A. Excessive voltage is supplied.

B. Excessive amperage is drawn. NOTE: Usually associated with a short circuit.



SUGGESTED DOK-LOK® MAINTENANCE

⚠ WARNING

A safe work place requires all lights and the horn to be working properly. DO NOT use DOK-LOK® if parts are broken or missing.

The lights can be tested by operating the DOK-LOK® and observing the light changes. The horn should sound whenever the hook is outside the ordinary work range. See pages 13 – 15.

DAILY

1. Remove debris around DOK-LOK®
2. Verify inside and outside lights and horn are working.
3. Replace damaged, missing light bulbs and lenses.
4. Repair, remount, or replace outside and inside signs as required.

90 DAYS

1. Do all Daily maintenance.
2. Spray lube chain.
3. Spray lube (anti-seize) the (6) carriage rollers.
4. Inspect dock bumpers. Replace if torn/worn. 4" of bumper protection is required.
5. Lubricate (anti-seize) limit switch mounting bracket between drive sprocket and cam.

180 DAYS

1. Do all Daily and 90 Day maintenance.
2. Verify that clutch torque setting is 575 in-lbs to 1000 in-lbs (48 – 85 ft-lbs). To adjust see Figure 13 and Clutch Adjustment on page 20.
3. Check and, if necessary, tighten concrete anchor bolts to 100 ft-lbs.

360 DAYS

1. Do all Daily, 90 Day, and 180 Day maintenance.

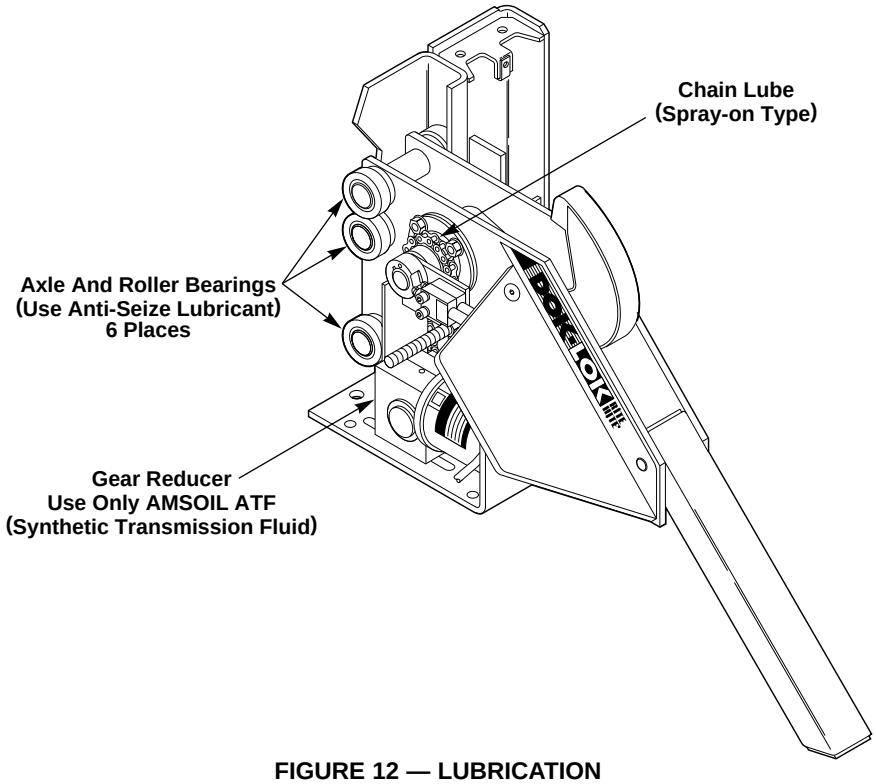
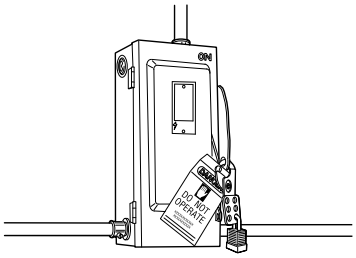


FIGURE 12 — LUBRICATION

2. Check and tighten, if necessary, sign and electrical box mounting bolts.
3. Check and tighten, if necessary, flexible electrical line box connections.
4. Inspect flexible electrical control harness from DOK-LOK® to junction box. Look for “kinks,” crushed areas, etc.
5. Check and tighten, if necessary, dock leveler limit switch mounting bolts (if equipped).
6. Check and tighten, if necessary, motor drive chain. To tighten see Figure 13.
7. Inspect outside junction box. Be sure it is rigidly mounted and the cover is tight with a gasket. If box is loose or “dented,” inspect all wires and wire connections.
8. Inspect the outside light-box. It should be rigidly mounted. If loose or damaged, inspect all wires and wire connections.

MAINTENANCE CHECK LIST

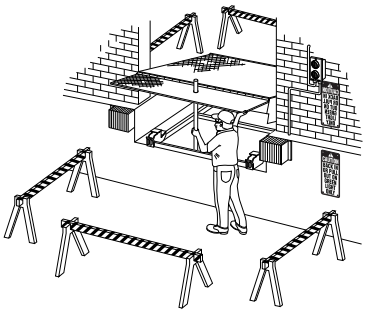
⚠ WARNING



LOCKOUT/TAGOUT PROCEDURES

When working with electrical or electronic controls make sure that the power source has been locked out and tagged according to OSHA regulation 1910.147 and approved local electrical codes.

⚠ WARNING



BARRICADE WORK AREA

Post safety warnings and barricade work area, at dock level and at ground level, to prevent unauthorized use of the dock position while service work is being performed.

Good housekeeping practices are the most commonly needed DOK-LOK® maintenance requirement. Shipping and receiving docks are notorious for having wood scraps, steel banding and other debris laying at the front of the dock.

ON A DAILY BASIS

NOTE

Test the DOK-LOK® using the following procedures. If required to repair or replace any item, lock out and tag out the power and barricade the dock position before proceeding.

1. Clean DOK-LOK® and dock area.
2. Test horn and all inside and outside lights.

⚠ WARNING

Do not use DOK-LOK® vehicle restraint if lights, horn, or other parts are broken or missing.

3. Replace burned out light bulbs as required.
4. Replace broken or missing red and green light cover lenses as required.
5. Repair, remount, or replace outside and inside signs as required.

IMPORTANT

DOK-LOK® must be protected by 4" thick bumpers or as required per your installation drawings. WORN, TORN, LOOSE, OR MISSING BUMPERS MUST BE REPLACED.

EVERY 90 DAYS

NOTE

If a leveler is installed at the LDL-500 DOK-LOK® location, it may be necessary to raise the leveler to remove the DOK-LOK® carriage assembly from the roller track. Use care when removing DOK-LOK® carriage assembly from roller track to not pull control harness wires out of wall mounted junction box. Raise the leveler, insert and secure the SAFE-T-STRUT™, and lock out and tag out the power source.

1. Inspect inside control box. If the box is hit by a forklift or material, bending damaging or loosening mountings and/or component pieces. Verify that electrical piping “in” and “out” of box is secure and not damaged.
2. Inspect flexible control harness from DOK-LOK® to junction box. Look for “kinks,” crushed areas, etc. Repair/replace as needed.
3. Inspect outside junction box. Be sure it is rigidly mounted and the cover is tight with a gasket. If box is loose or “dented,” inspect all wires and wire connections and repair/replace as needed.
4. Inspect the outside light-box. It should be rigidly mounted. If loose or dented, inspect all wires and wire connections and repair/replace as needed.

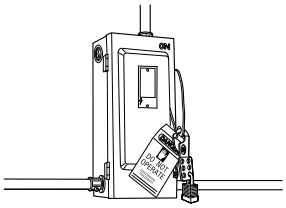
⚠ CAUTION

Use proper lifting device when removing or lifting the LDL-500 from the roller track. Carriage weight is approximately 110 lbs.

5. Clean and lubricate the six (6) roller bearings on the carriage assembly. Remove the two spring covers and remove the lower spring bar mounting bolt nuts. Using an appropriate lifting device, remove the carriage assembly from the roller track plate.
6. Clean the six (6) roller bearings and lubricate using an anti-seize type of lubricant. Clean the roller track plate making sure to wipe out any grease and dirt along the entire length of both sides of the track.
7. Using an appropriate lifting device, reinstall the DOK-LOK® carriage assembly and attach the springs, spring cover plates and lower spring bar mounting bolts.
8. Perform operational test after all maintenance repairs and adjustments are completed.

CLUTCH ADJUSTMENT

⚠ WARNING



LOCKOUT/TAGOUT PROCEDURES

When working with electrical or electronic controls make sure that the power source has been locked out and tagged according to OSHA regulation 1910.147

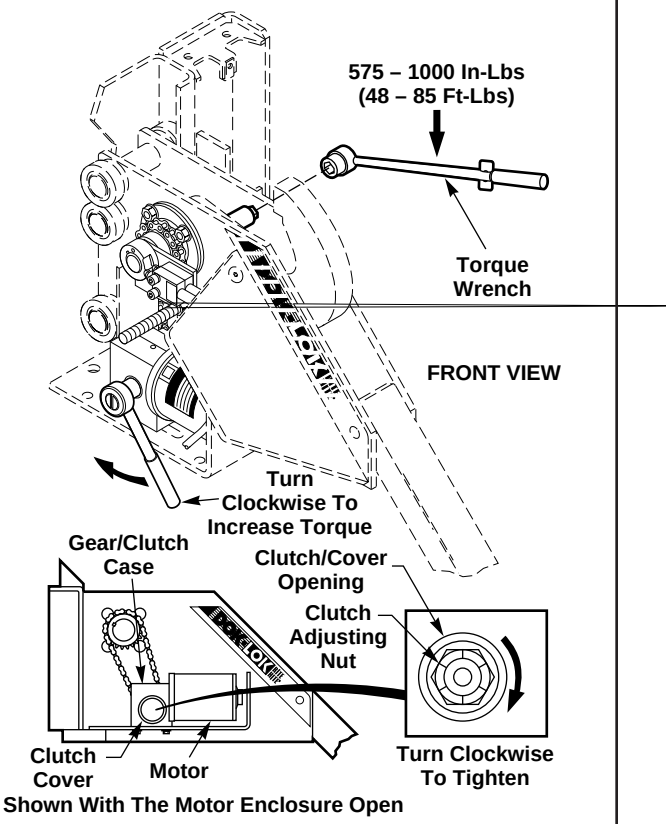


FIGURE 13 — CLUTCH ADJUSTMENT

Open the motor enclosure. Pry off the clutch cover to expose the clutch adjusting nut. Refer to Figure 13.

NOTE

Clutch adjustments are to be made with hook at a 90° hooked position. (See Figure 14.) This is a must for proper adjustment.

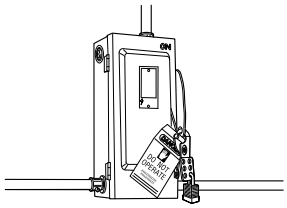
New DOK-LOK® may have higher torque readings. Do not adjust downward.

To tighten the clutch (reduce slippage) turn the clutch adjustment nut clockwise. Using an inch-pound torque wrench on the axle side for measuring the torque, continue to turn the clutch adjusting nut until 700 in-lbs of torque is obtained on the shaft end. Refer to Figure 13. Use a dial indicator or bridge type inch-pound torque wrench to properly torque nut. Do not use a clicker type torque wrench.

Put a coating of silicone sealer on the clutch cover, then reinstall the clutch cover and close the motor enclosure.

DRIVE CHAIN ADJUSTMENT

⚠ WARNING



LOCKOUT/TAGOUT PROCEDURES

When working with electrical or electronic controls make sure that the power source has been locked out and tagged according to OSHA regulation 1910.147

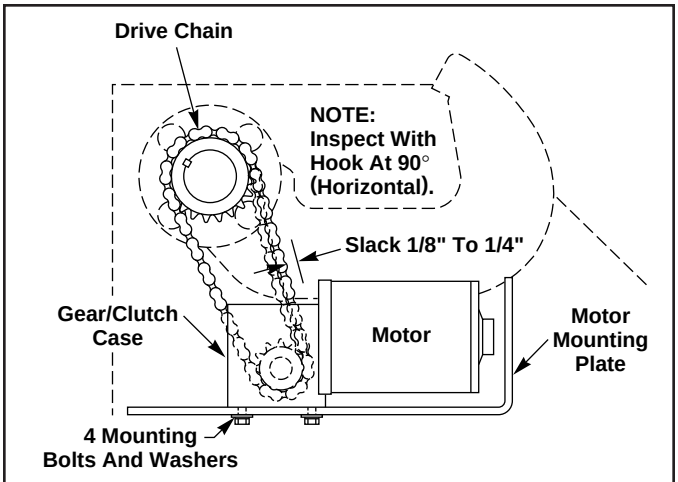
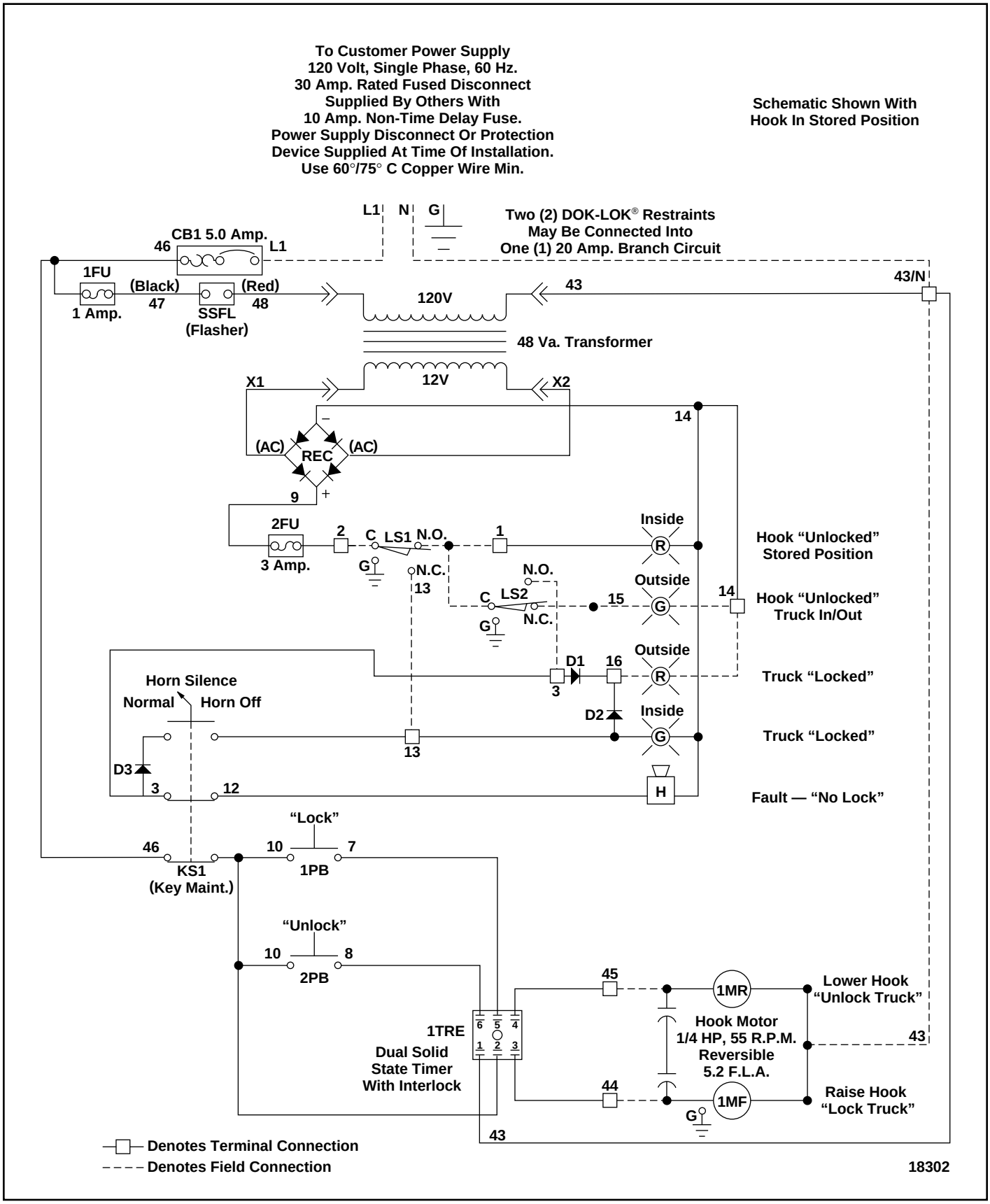


FIGURE 14 — DRIVE CHAIN ADJUSTMENT

Periodically inspect the drive chain for dirt and chain "slack." Open the motor enclosure for inspection. Clean the chain with solvent. After cleaning, spray the chain with a good quality chain spray type lubricant.

The drive chain should NOT HAVE MORE THAN 1/4" SLACK. Refer to Figure 14. To tighten chain, open motor enclosure, loosen the 4 motor mounting bolts. Pry the motor assembly forward, in the slotted bolt holes, until the chain is tight with proper alignment between sprockets. Hold in this position and tighten the 4 motor mounting bolts to 105 in-lbs in increments of 10 in-lbs. Tighten in ⤫ pattern. Test for proper chain alignment. Close the motor enclosure. Turn on power.

ELECTRICAL SCHEMATIC

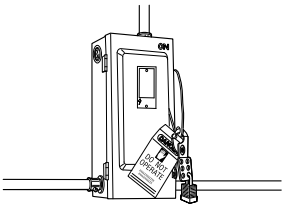


TROUBLESHOOTING

⚠ WARNING

To prevent personal injury use extreme care when working with electrically powered equipment. Follow all safety instructions. Disconnect all power, and keep body away from moving parts. Use caution with compressed air; foreign materials can be blown through the skin or into the eyes. Use only OSHA approved air nozzles and do not exceed recommended air pressure.

⚠ WARNING



LOCKOUT/TAGOUT PROCEDURES

When working with electrical or electronic controls make sure that the power source has been locked out and tagged according to OSHA regulation 1910.147

Problem	Probable Cause	Solution
1. DOK-LOK® hook does not raise and lights do not flash.	Power source is not functional.	Check building circuit breaker and the circuit breaker in the control box.
	Improper wiring.	Check wiring by tracing out entire circuit. Refer to electrical schematic inside control box enclosures.
2. DOK-LOK® lights are flashing but hook does not raise/lower.	Chain is loose or broken.	Repair or replace as required.
	Electrical malfunction.	Check wiring connections at the inside control box, the outside junction box and the DOK-LOK® motor cover box. Use volt/ohmmeter to verify wiring.
	Drive motor defective.	Repair or replace as required.
	Timer defective.	Repair or replace as required.
	LOCK (raise) or UNLOCK (lower) pushbutton switch defective.	Repair or replace as required.
	Keyswitch is in “HORN OFF” position.	Turn keyswitch to “normal” position.

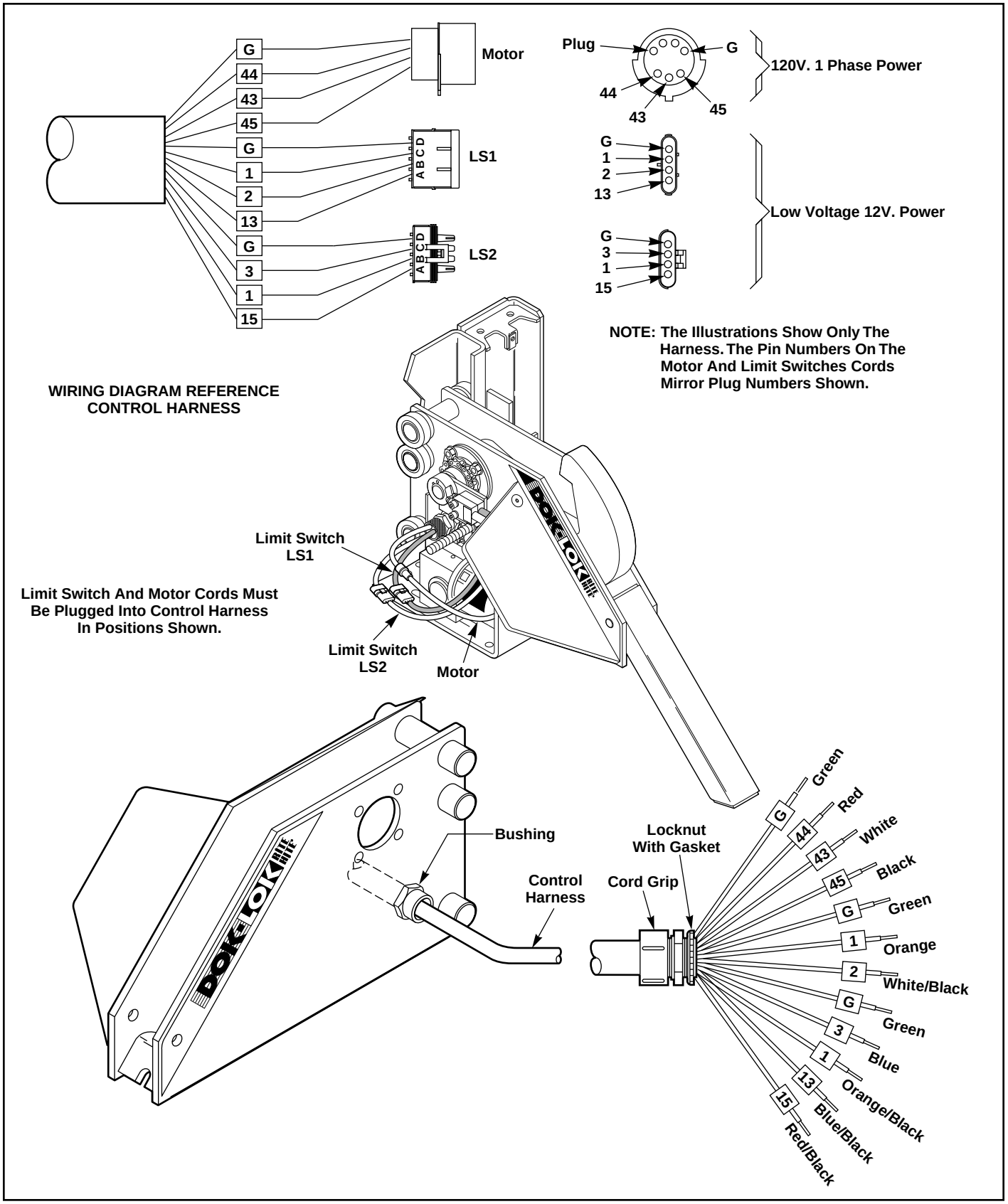
Problem	Probable Cause	Solution
3. DOK-LOK® lights are flashing but hook does not raise/lower to full extent.	Drive motor clutch out of adjustment.	Refer to page 19 for clutch adjustment procedures.
	Drive motor mounting bolts coming loose.	Check alignment between motor sprocket, roller chain, and hook shaft sprocket. Torque bolts to 105 in-lbs.
	Low incoming voltage.	Power/voltage supply must be a minimum of 500 VA, if source is a control transformer (i.e., an overhead door operating motor).
4. DOK-LOK® is operational, but hook drops, causing lights to change and/or horn to sound while trailer is being serviced.	Drive motor clutch out of adjustment.	Refer to page 19 for clutch adjustment procedures.
5. Hook is operational but all lights are out.	Blown fuse.	Check 3 amp line fuse inside control box.
	Bulbs burned out, loose, or missing.	Check all bulbs and replace as required.
	Improper wiring connections.	Check all wiring connections. A good ground is required. Ask customer if anyone attempted repairs.
	Defective transformer.	Check for defective transformer. Failures often are noticed by “melted wires” and a burnt smell.
	Defective flasher.	Flasher’s contacts can fail open or closed. If closed, you will see a solid red or green light. If open, you will see no lights.
	Defective bridge rectifier.	Replace as required.
6. Lights do not change from red to green (or vice versa), but hook and horn are operational.	Improper wiring.	Verify wiring.
	Defective LS2 limit switch.	Check for defective limit switch, LS2. Disconnect power before testing limit switches.
7. Horn does not sound but lights and hook are operational.	Improper wiring.	Check all wiring connections.
	Horn is not working properly.	Check and test alarm horn. For testing, momentarily touch a jumper wire between wire no. 2 on the terminal strip (inside control box) to no. 12 wire connection at the horn. The horn will sound if operational.
	Defective limit switch.	Replace as required.

Problem	Probable Cause	Solution
8. Horn sounds continuously when lights and hook are operational.	Improper wiring.	Check all wiring connections.
	Defective limit switch LS2.	Replace as required.
9. DOK-LOK® carriage does not return to a full up position.	Carriage binding in track.	Check to see if roller track plate is clean and rollers are clean, free of debris, and lubricated. Use only an anti-seize lubricant to lubricate rollers.
	Damaged roller track plate.	Verify that the roller track plate is straight and not damaged.
	Broken or weak springs.	Remove spring cover and replace as required.

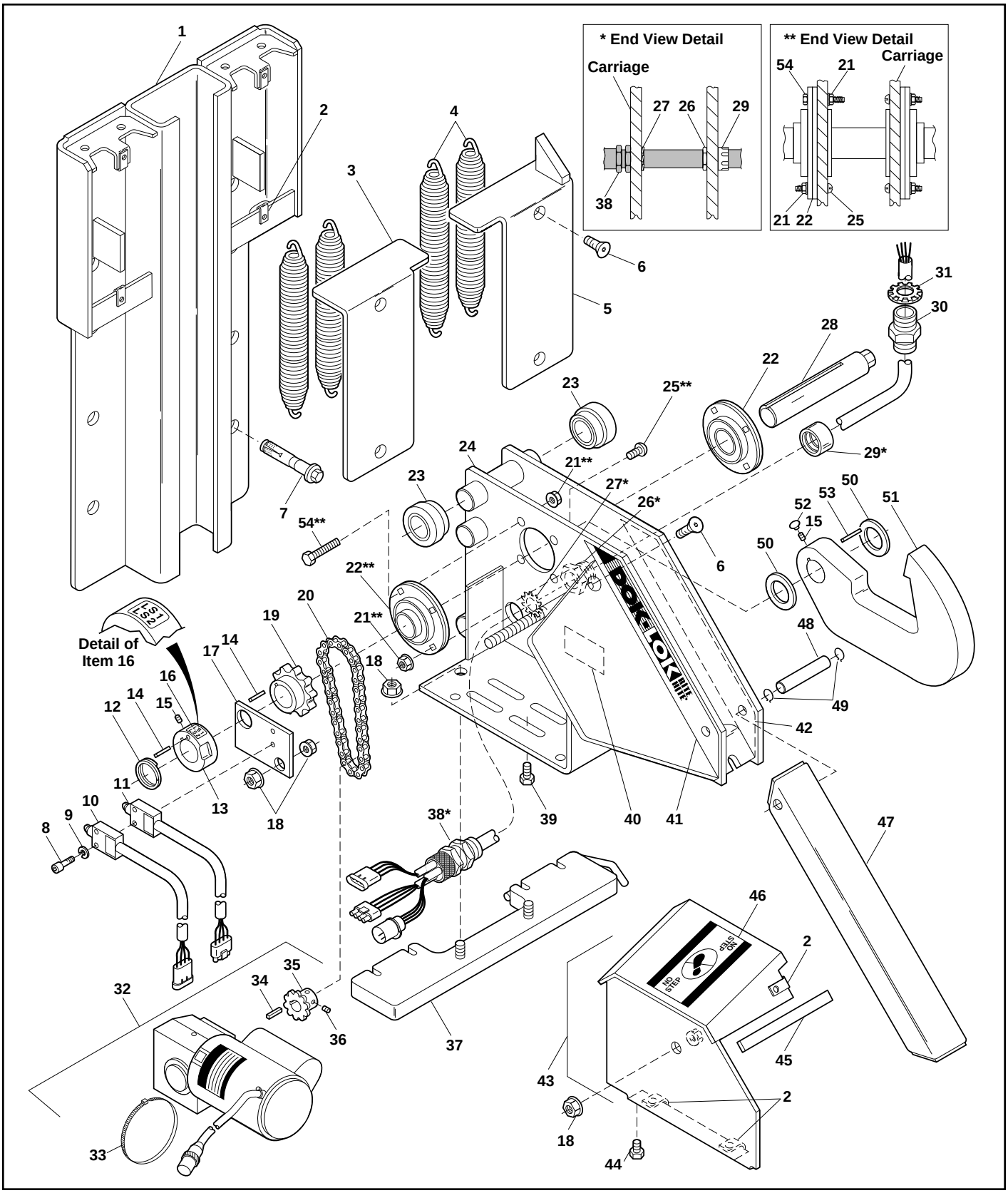
COMPONENT TESTING CHART

Component	Test	Result
A. Timer Test Procedure — Time #55790		Manually pull hook up, or run hook up electrically to “locked” position (90° to dock face) and turn power off. Turn power on and observe hook. • If hook goes down immediately, “unlock” section of timer is failed closed. • If hook goes up immediately, “lock” section of timer is failed closed. • If hook does not move, push LOCK button. If hook still does not move, turn off power, remove wires #7 and #44 from timer and tape together, push LOCK button. If the motor runs as long as the button is depressed, the “lock” section of the timer is failed open. • Repeat the previous procedure with wires #8 and #45 to test “unlock” section. Replace timer if any of the above failed conditions exist.
B. Motor Testing for Faults — Electrical with V.O.M. Meter (to be done by a qualified electrician).	Infinite ohms (no needle movement) between M1 and M2.	Check between 44 and 43 or 45 and 43 to determine which winding is open.
	Mechanical Bind (locked rotor). Motor leads show continuity between all windings. Shaft does not move. Reads over 8 amps at 43 on both rotations.	Determine reason for binding and repair or replace.
C. Limit Switch Test	Plunger depressed.	Continuity between terminals C and B.
D. Limit Switch Test	Plunger in “free state.”	Continuity between terminals A and B.

CONTROL HARNESS



REPLACEMENT PARTS – STANDARD

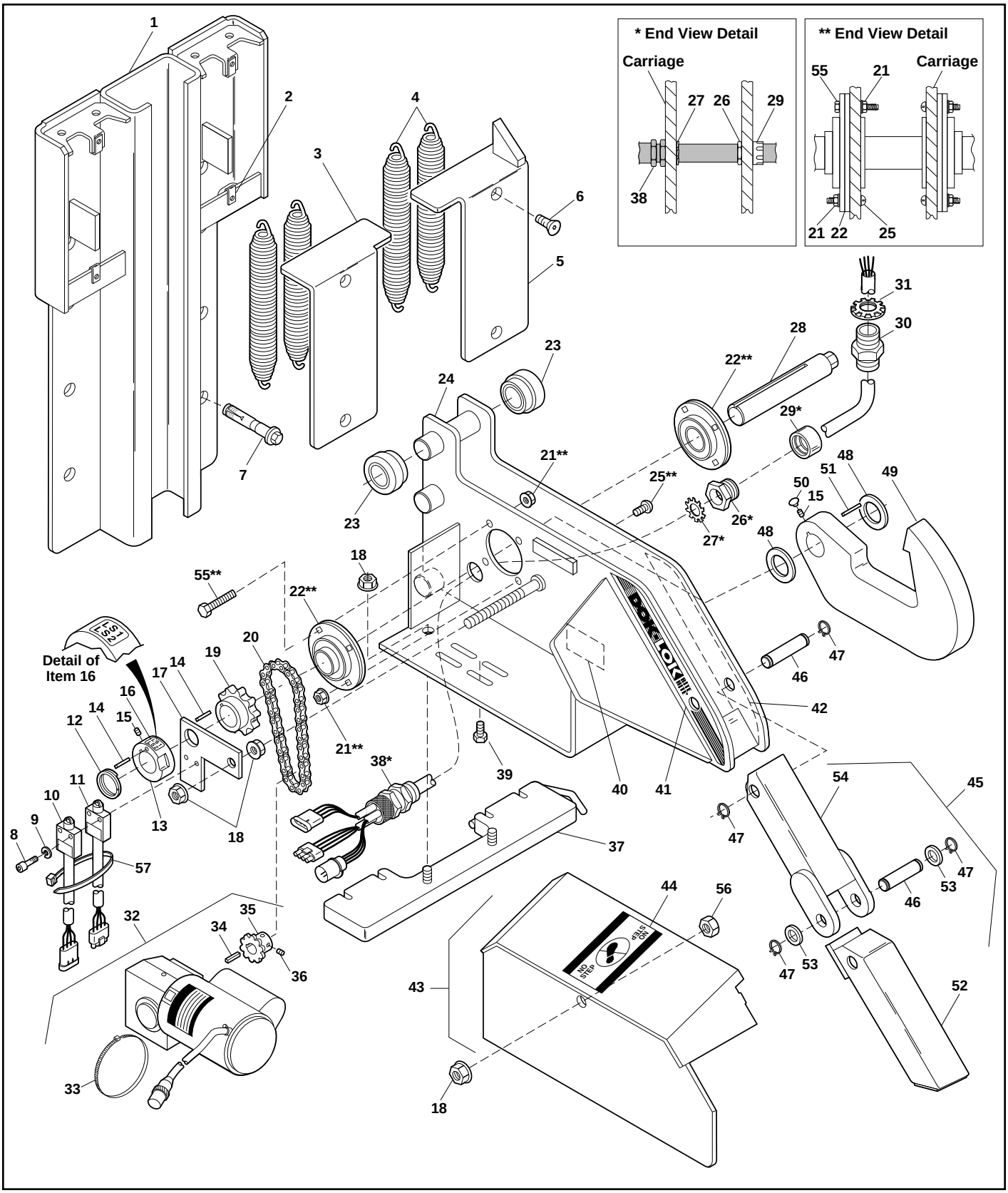


Pub. No. 1139-R3 — March 1999

REPLACEMENT PARTS LIST – STANDARD

Item	Quantity	Description	Part Number
1	1	Roller Track	1955
2	7	Cage Nut	56540
3	1	Spring Cover Left Hand	54849
4	4	Extension Spring	52129
5	1	Spring Cover Right Hand	1954
6	5	Mounting Bolt (5/16" – 18NC x 1" Long) Flat Head Socket	18210
7	15	Concrete Anchor (5/8" x 4" Hex Head) – Includes Bolt and Washer	53150
8	2	Socket Head Screw (1/4" – 20NC x 1-1/2" Long)	51636
9	2	Hi-Collar Lockwasher (1/4")	51802
10	1	LS2 Limit Switch (Includes Item 33)	18212
11	1	LS1 Limit Switch (Includes Item 33)	18211
12	1	Spiral Retaining Ring (1-3/16" ID)	56526
13	1	Cam	53432
14	2	Sprocket and Cam Key (1/4" x 1/4" x 7/8" Long)	54864
15	2	Sprocket, Cam and Hook Set Screw (5/16" – 18NC x 5/16" Long)	56525
16	1	Cam Decal	18213
17	1	Limit Switch Mounting Plate	18285
18	6	Serrated Flange Whiz Nut (7/16" – 14NC)	51549
19	1	Hook Drive Sprocket with Set Screw (#35 – 27 Tooth)	56177
20	1	Chain	56175
21	8	Serrated Flange Whiz Nut (3/8" – 16NC)	51564
22	2	Bearing	6085
23	6	Carriage Roller	15782
24	1	Carriage	1973
25	7	Carriage Bolt (3/8" – 16NC x 1" Long)	58022
26	1	Conduit Nipple	18204
27	1	Locknut	55791
28	1	Hook Shaft	54987
29	1	Conduit Bushing	57978
30	1	Harness Cord Grip (3/4" x 11/16" – 13/16")	66082
31	1	Harness Locknut with Gasket	55877
32	1	Motor Assembly (Includes Items 33, 34, 35 and 36)	18207
33	1	Wire Tie (Motor and Limit Switch)	18548
34	1	Sprocket Key	18221
35	1	Sprocket (#35 – 10 Tooth)	56176
36	1	Set Screw (1/4" – 20NC x 3/16" Long)	68090
37	1	Lower Spring Bar Mounting Plate	1957
38	1	Control Harness (Item 27 is Locknut for Harness)	18205
39	4	Bolt (5/16" – 18NC x 5/8" Long)	58069
40	1	Patent Number Tag	18391
41	1	Carriage Logo Decal Left Side	59009
42	1	Carriage Logo Decal Right Side	59008
43	1	Motor Cover (Complete Assembly Includes Items 2 [Qty. 3], 45 and 46)	1995
44	2	Motor Cover Bolt (5/16" – 18NC x 1" Long)	58017
45	1	Foam Water Seal	56542
46	1	No Step Warning Decal	54126
47	1	Slope Extension	1883
48	1	Slope Extension Pivot Pin	52673
49	2	Retaining Ring (5/8" ID)	56775
50	2	Machinery Bushing	51745
51	1	Hook	18208
52	1	Protective Plug	56523
53	1	Hook Key (1/4" x 1/4" x 1-1/4" Long)	54818
54	1	Bolt Hook Stop	68207

REPLACEMENT PARTS – 9" CARRIAGE OPTION

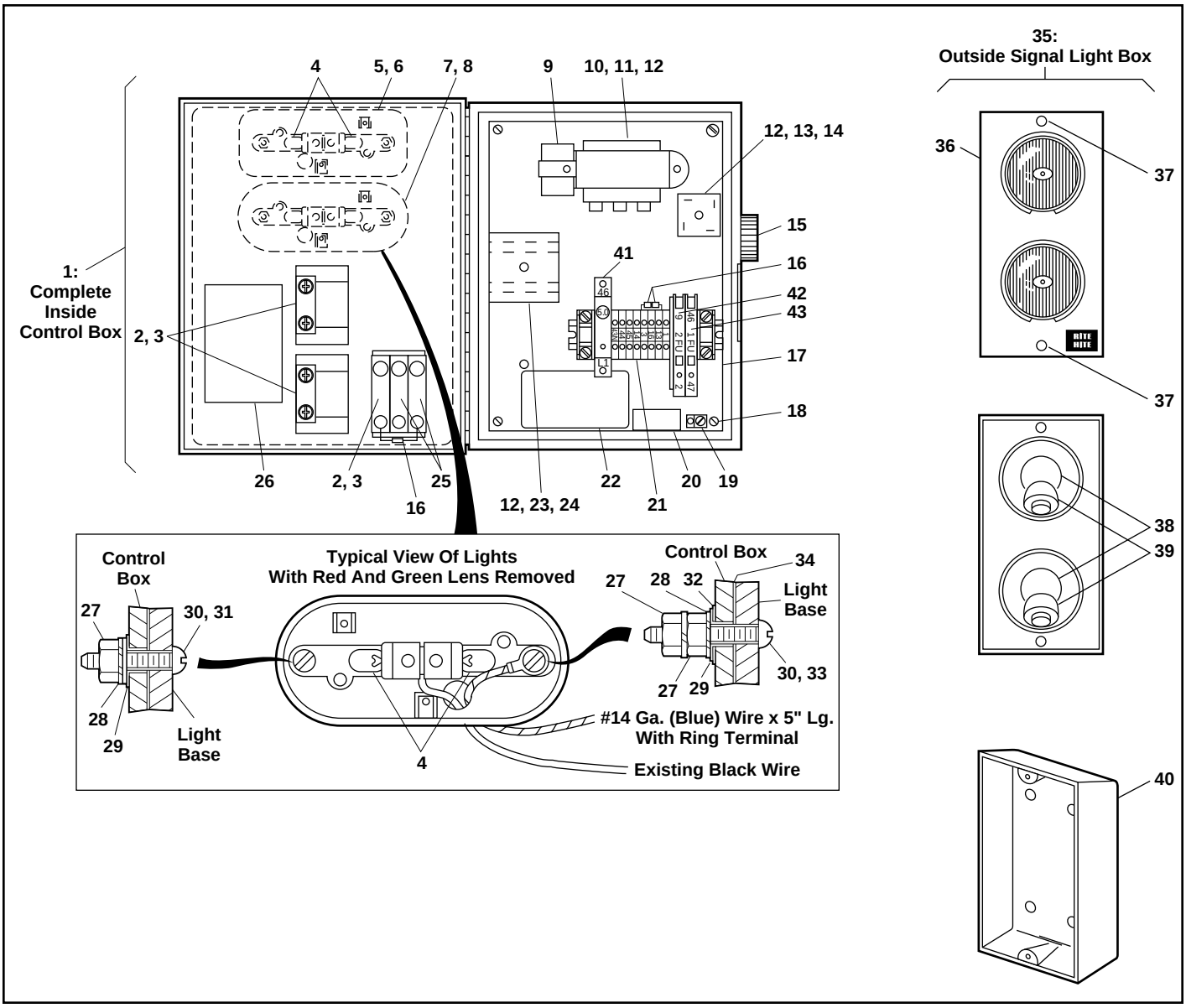


Pub. No. 1139-R3 — March 1999

REPLACEMENT PARTS LIST – 9" CARRIAGE OPTION

Item	Quantity	Description	Part Number
1	1	Roller Track	1955
2	4	Cage Nut	56540
3	1	Spring Cover Left Hand	54849
4	4	Extension Spring	52129
5	1	Spring Cover Right Hand	1954
6	4	Mounting Bolt (5/16" – 18NC x 1" Long) Flat Head Socket	18210
7	15	Concrete Anchor (5/8" x 4" Hex Head) – Includes Bolt and Washer	53150
8	2	Socket Head Screw (1/4" – 20NC x 1-1/2" Long)	51636
9	2	Hi-Collar Lockwasher (1/4")	51802
10	1	LS2 Limit Switch (Includes Item 33)	18212
11	1	LS1 Limit Switch (Includes Item 33)	18211
12	1	Spiral Retaining Ring (1-3/16" ID)	56526
13	1	Cam	19123
14	2	Sprocket and Cam Key (1/4" x 1/4" x 7/8" Long)	54864
15	2	Sprocket, Cam and Hook Set Screw (5/16" – 18NC x 5/16" Long)	56525
16	1	Cam Decal	18213
17	1	Limit Switch Mounting Plate	19124
18	5	Serrated Flange Whiz Nut (7/16" – 14NC)	51549
19	1	Hook Drive Sprocket with Set Screw (#35 – 27 Tooth)	56177
20	1	Chain	19125
21	8	Serrated Flange Whiz Nut (3/8" – 16NC)	51564
22	2	Bearing	6085
23	6	Carriage Roller	15782
24	1	Carriage	18945
25	7	Carriage Bolt (3/8" – 16NC x 1" Long)	58022
26	1	Conduit Nipple	18204
27	1	Locknut	55791
28	1	Hook Shaft	54987
29	1	Conduit Bushing	57978
30	1	Harness Cord Grip (3/4" x 11/16" – 13/16")	66082
31	1	Harness Locknut with Gasket	55877
32	1	Motor Assembly (Includes Items 33, 34, 35 and 36)	18207
33	1	Wire Tie (Motor)	18548
34	1	Sprocket Key	18221
35	1	Sprocket (#35 – 10 Tooth)	56176
36	1	Set Screw (1/4" – 20NC x 3/16" Long)	68090
37	1	Lower Spring Bar Mounting Plate	1957
38	1	Control Harness (Item 27 is Locknut for Harness)	18205
39	4	Bolt (5/16" – 18NC x 5/8" Long)	58069
40	1	Patent Number Tag	18391
41	1	Carriage Logo Decal Left Side	19031
42	1	Carriage Logo Decal Right Side	19030
43	1	Motor Cover (Complete Assembly Includes Item 44)	19213
44	1	No Step Warning Decal	54126
45	1	Slope Extension – Assembly (Includes Items 46, 47 [Qty. 2], 52, 53 [Qty. 2] and 54)	19214
46	2	Slope Extension Pivot Pin (1" x 3-1/2")	19126
47	4	Retaining Ring (1" ID)	19127
48	2	Machinery Bushing – Hook	51745
49	1	Hook	18208
50	1	Protective Plug	56523
51	1	Hook Key (1/4" x 1/4" x 1-1/4" Long)	54818
52	1	Slope Extension – Lower	18960
53	2	Machinery Bushing (.06 thick x 1.015 ID)	19128
54	1	Slope Extension – Upper	18961
55	1	Bolt Hook Stop	68207
56	1	Ny-Lock Nut (7/16" – 14NC)	51543
57	1	Wire Tie (Limit Switch)	55592

REPLACEMENT PARTS

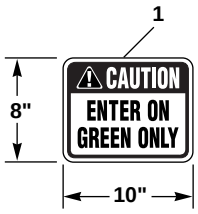


Pub. No. 1139-R3 — March 1999

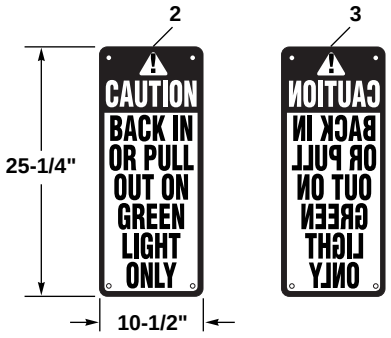
REPLACEMENT PARTS LIST

Item	Required	Description	Part Number
1	1	Control Box Assembly (Complete)	18301
2	3	Contact Block — 1 N.O.	65890
3	3	Mounting Latch	65889
4	4	Light Bulb #194	55768
5	1	Inside Signal Light Red Assembly	57803
6	1	Inside Signal Light Red Lens	57840
7	1	Inside Signal Light Green Assembly	55742
8	1	Inside Signal Light Green Lens	55812
9	1	Flasher, Solid State — 115V AC	55733
10	1	Transformer	57327
11	4	Round Head Screw #8-32 x 1/2" Long	51699
12	6	#8 Star Washer	51760
13	1	Bridge Rectifier — 3A., 50 P.I.V.	55760
14	1	Round Head Screw #8-32 x 3/4" Long	51698
15	1	Horn — 12V.	57383
16	3	Diode — 10A., 50 P.I.V.	55761
17	1	Sub-Panel	8318
18	4	Round Head Screw (#10-32 x 3/8" Long)	58193
19	1	Ground Lug	55902
20	1	Ground Lug Data Decal	54069
21	1	Terminal Block (With Fuse and Circuit Breaker)	18329
22	1	Fuse Replacement Label	64216
23	1	Round Head Screw #8-32 x 1" Long	51689
24	1	2-Stage, Dual Timer	55790
25	2	Contact Block — 1 N.C.	65896
26	1	Full Load Amp and Voltage Decal	18365
27	6	Hex Nut #8-32	51538
28	4	Lockwasher #8	51760
29	4	Nylon Bushing	53147
30	2	Round Head Screw #8-32 x 3/4" Long	51672
31	1	Round Head Screw #8-32 x 1/2" Long (Green Light Only)	51645
32	2	Nylon Washer	53164
33	1	Round Head Screw #8-32 x 1" Long (Red Light Only)	51656
34	1	Gasket (Red Light Only)	65262
35	1	Outside Light Box (Complete)	18280
36	1	Outside Light Box Cover (With Lamp Assemblies and Lens)	18278
37	2	Round Head Screw #6-32 x 3/4" Long	18364
38	2	Light Bulb (GE #1156)	57187
39	2	Lamp Socket Assembly	57186
40	1	Outside Light Box, NEMA 3R (Without Cover)	18277
41	1	Circuit Breaker 5.0 Amp	65609
42	1	3.0 Amp Slo-Blo Fuse	55784
43	1	1.0 Amp Slo-Blo Fuse	57428

REPLACEMENT PARTS



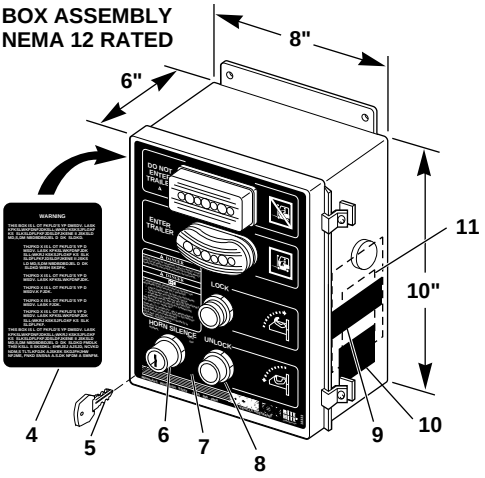
INSIDE SIGN

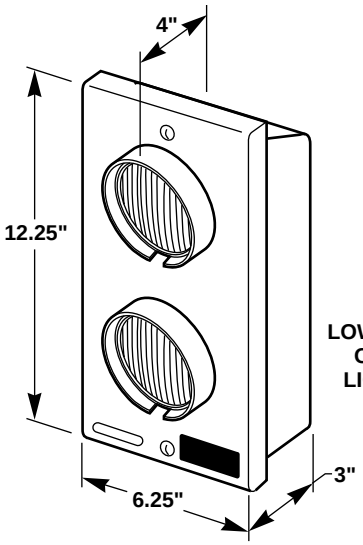


OUTSIDE SIGNS

OVERALL DIMENSION VIEW

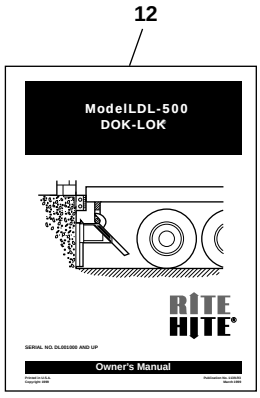
INSIDE CONTROL BOX ASSEMBLY NEMA 12 RATED





LOW PROFILE OUTSIDE LIGHT BOX

OWNER'S MANUAL



Item	Required	Description	Part Number
1	1	Interior Warning Sign	56095
2	1	Narrow Sign (Normal Letters)	56112
3	1	Narrow Sign (Reverse Letters)	56113
4	1	Warning Decal	18358
5	1	Key (Replacement)	65897
6	1	Keyswitch — 2 Position	65899
7	1	LDL Control Box Decal	18333
8	2	Pushbutton	65887
9	1	Motor Reset Decal	54051
10	1	Lockout/Tagout Decal	59170
11	1	Schematic Decal	59003
12	Additional Copies	Owner's Manual	Pub. No. 1139-R3

RITE-HITE® STANDARD WARRANTY

RITE-HITE® warrants that its products will be free from defects in design, materials, and workmanship for a period of 365 days from the date of shipment. All claims for breach of this warranty must be made within 30 days after the defect is or can, with reasonable care, be detected and in no event no more than 30 days after the warranty has expired. In order to be entitled to the benefits of this warranty, the products must have been properly installed, maintained, and operated within their rated capacities and/or specified design parameters, and not otherwise abused. Periodic lubrication and adjustment is the sole responsibility of the owner. This warranty is RITE-HITE®'s exclusive warranty. RITE-HITE® EXPRESSLY DISCLAIMS ALL IMPLIED WARRANTIES, INCLUDING THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS. Non-standard warranties, if any, must be specified by RITE-HITE® in writing.

In the event of any defects covered by this warranty, RITE-HITE® will remedy such defects by repairing or replacing any defective equipment or parts, bearing all the costs for parts, labor, and transportation. This shall be the exclusive remedy for all claims whether based on contract, negligence, or strict liability.

LIMITATION OF LIABILITY

RITE-HITE® SHALL NOT IN ANY EVENT BE LIABLE FOR ANY LOSS OF USE OF ANY EQUIPMENT OR INCIDENTAL OR CONSEQUENTIAL DAMAGES OF ANY KIND, WHETHER FOR BREACH OF WARRANTY, NEGLIGENCE, OR STRICT LIABILITY.

RITE HITE®

National Sales & Service Office:

RITE-HITE®
8900 N. Arbon Drive
P.O. Box 23043
Milwaukee, Wisconsin 53223

Phone: 414-355-2600

Representatives in all Major Cities